

# Risk Sharing, Relative Position, and Position-Dependent Enforcement in Village Economies

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## Abstract

I study the role of relative position in informal risk sharing in Indian village economies. Using household panel data from three ICRISAT villages, I first document a robust pattern: a household's rank in the within-group distribution, whether measured through consumption or income, is a powerful predictor of consumption dynamics, significant at the 1% level across all villages and castes. Controlling for rank substantially reduces or eliminates rejections of efficient risk sharing. The finding holds with both consumption rank (motivated by a preference-based derivation) and income rank, which is free of mechanical correlation with contemporaneous consumption changes and provides a direct bridge to the structural model. To quantify the structural importance of this finding, I embed position-dependent enforcement, where social sanctions upon deviation depend on a household's income-state rank, into [Bold and Broer's \(2021\)](#) limited commitment model with coalitional deviations. Estimating via simulated method of moments, I find the data consistently prefer position-dependent enforcement ( $\omega = 0.95$  across all three villages), with the SMM criterion improving by 46–56% relative to the baseline. The improvement is driven primarily by cross-sectional consumption dispersion: the model matches 51–82% of observed dispersion across villages, compared to 14–30% in the baseline, while preserving the core findings of small stable groups and approximately symmetric insurance.

## 1 Introduction

In developing countries, the welfare of rural households depends on their ability to smooth consumption against idiosyncratic shocks. Despite the absence of formal insurance markets, households use existing institutions such as gifts, transfers, loans, and marriage to pool risk informally ([Rosenzweig and Stark, 1989](#); [Townsend, 1994](#)). A large literature tests the efficiency of these arrangements and, when full insurance is rejected, seeks to identify the frictions that limit risk sharing.

Two developments in the literature motivate this paper. First, [Bold and Broer \(2021\)](#) show that the standard limited commitment model with individual deviations predicts counterfactually large asymmetry in consumption responses in village-sized groups. Allowing for coalitional

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deviations as in [Genicot and Ray \(2003\)](#) endogenously produces small stable groups of 4 households, which generate approximately symmetric insurance that better matches the data. Second, a growing body of evidence establishes that relative deprivation affects subjective welfare and economic behaviour: [Luttmer \(2005\)](#) shows that neighbours’ earnings reduce own well-being in the United States, and [Fafchamps and Shilpi \(2008\)](#) document similar effects of relative consumption in Nepal. More broadly, relative position within a reference group predicts consumption, transfers, and migration decisions in developing countries ([Stark and Taylor, 1991](#); [Gortner and van der Weele, 2019](#); [Pattath, 2020](#); [Arcand and Pattath, 2023](#)).

A natural connection links these two strands. In the risk-sharing literature, the relevant group is typically defined by spatial proximity and shared characteristics such as village, lineage, caste, or ethnicity because these features enable monitoring and enforcement that help bypass asymmetric information problems ([Genicot and Ray, 2003](#)). But these same factors also define the social reference groups to which households compare their incomes, consumption, and well-being. If risk-sharing groups and social comparison groups substantially overlap, then relative deprivation inevitably arises among households that simultaneously share risk and compare themselves to one another. This is especially plausible in rural India, where within-caste interactions, endogamy ([Rosenzweig and Stark, 1989](#); [Munshi and Rosenzweig, 2009](#)), and conspicuous consumption through marriage expenditures ([Bloch et al., 2004](#)) create tight reference groups. The within-group inequality and consequent relative deprivation felt by households could therefore affect the quantity and direction of gifts, transfers, and loans, as well as the decision to form familial associations through marriage ([Linssen et al., 2011](#)). Yet the risk-sharing literature has not asked whether relative position also affects the sustainability of informal insurance.

I bridge these two strands in two steps. In the first step, I augment the standard [Townsend \(1994\)](#) risk-sharing test with measures of relative position, specifically, a household’s rank in the empirical cumulative distribution function within its risk-sharing group. Using household panel data from the ICRISAT Village Level Studies for three Indian villages, I show that rank, measured through either consumption or income, is a predictor of consumption dynamics, significant at the 1% level across all villages and castes. Controlling for rank substantially reduces or eliminates rejections of efficient risk sharing. I present results with both consumption rank (directly motivated by a preference-based derivation) and income rank, which is free of the mechanical correlation between contemporaneous consumption rank and consumption changes and roughly doubles the  $R^2$  of the standard specification.

This reduced-form evidence establishes that income position matters for risk-sharing outcomes, but it is agnostic about the underlying mechanism. Households may care directly about their relative consumption (a preference channel), or communities may enforce risk-sharing norms through sanctions that depend on a household’s economic standing (an enforcement channel), or both.

In the second step, I discipline this finding structurally. I extend [Bold and Broer \(2021\)](#)’s limited commitment model with coalitional deviations by introducing position-dependent enforcement of insurance arrangements. When a household deviates from the risk-sharing group, the community imposes social sanctions such as exclusion, loss of social capital, reduced access

to common resources and the severity of these sanctions depends on the household’s income position. Specifically, low-income households face harsher punishment for defection than high-income households. This captures the empirically grounded idea that poorer households are more vulnerable to social punishment because they have less bargaining power and depend more heavily on community goodwill (Fafchamps, 2011).

The structural model adds a single parameter,  $\omega$ , which governs the intensity of position-dependent sanctions. When  $\omega = 0$ , I recover Bold and Broer (2021)’s baseline model exactly. Estimating  $\omega$  jointly with preference parameters via simulated method of moments (SMM) on seven data moments, I find:

- The data consistently prefer position-dependent enforcement:  $\omega = 0.95$  is selected across all three villages, with the SMM criterion improving by 46–56% over the baseline. The direction of this preference is robust; however,  $\omega$  is at the boundary of my grid in all villages, and the criterion declines monotonically in  $\omega$ , so the precise magnitude is not sharply identified.
- The model substantially reduces Bold and Broer’s under-prediction of cross-sectional consumption dispersion, matching 55–90% of the data target across villages versus 14–30% in the baseline.
- Core findings are preserved: stable groups of 3–4 households as in Bold and Broer (2021), approximately symmetric insurance, and reasonable preference parameter estimates, with the discount factor estimated in the interior of its grid (0.96–0.97).
- The mechanism is robust to functional form: linear position-dependent sanctions dominate a convex alternative, suggesting the data prefer maximum differentiation between income positions.

Like Bold and Broer (2021), my model generates near-zero asymmetry in insurance responses, falling short of the negative asymmetry observed in the data. I discuss why limited commitment models with CRRA utility are structurally constrained in generating large asymmetries, and suggest directions such as loss aversion or asymmetric monitoring that may address this open puzzle.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 presents the reduced-form evidence on relative position and risk sharing. Section 4 develops the structural model with position-dependent enforcement. Section 5 presents the estimation strategy and results. Section 6 discusses the findings, and Section 7 concludes.

## 2 Related Literature

My paper contributes to three literatures: risk sharing in village economies, limited commitment models with endogenous groups, and the role of social comparisons in economic decisions. I survey each in turn, locating my contribution relative to the closest existing work.

## 2.1 Risk sharing in village economies

The theoretical foundations of risk sharing date to [Wilson \(1968\)](#), whose syndicates model established that under complete markets and common beliefs, efficient risk sharing implies that more risk-averse agents bear less risk. [Townsend \(1994\)](#) developed the canonical empirical test: under full insurance within a reference group, household consumption should co-move one-for-one with group-average consumption, and idiosyncratic income should have no additional predictive power. Applying this test to household panel data from three ICRISAT villages in semi-arid India, [Townsend \(1994\)](#) found that while the full insurance hypothesis is statistically rejected, the model provides a “surprisingly good benchmark” (see also [Townsend, 1995](#); [Deaton, 1990](#); [Ravallion and Chaudhuri, 1997](#)).

Since this benchmark, a large literature has documented systematic deviations from full insurance and sought to identify their sources. [Morduch \(2004\)](#) and [Mazzocco and Saini \(2012\)](#) revisit the ICRISAT data and show that inference is sensitive to the choice of testing specification and the treatment of preference heterogeneity: when preferences are allowed to differ across households, some rejections of efficient risk sharing prove spurious. [Chiappori et al. \(2014\)](#) provide formal conditions under which heterogeneous risk preferences can be distinguished from incomplete insurance. [Kinnan \(2022\)](#) disentangles limited commitment, moral hazard, and hidden income as distinct barriers to insurance in Thai villages, finding that limited commitment is the dominant friction. [Morten \(2019\)](#) demonstrates a trade-off between informal risk sharing and temporary migration in the ICRISAT villages: when migration opportunities expand, the returns to within-village risk sharing decline, and households partially substitute geographic diversification for group insurance.

A recurring finding in this literature is that the village may not be the relevant unit of risk sharing. [Grimard \(1997\)](#) shows that ethnicity, not geography, best predicts consumption smoothing in Côte d’Ivoire. For India, several studies converge on caste as the operative risk-sharing group ([Munshi and Rosenzweig, 2009](#); [Mazzocco and Saini, 2012](#); [Munshi, 2019](#)). [Munshi \(2019\)](#) argues that the persistence of caste-based networks reflects their insurance value: caste networks enforce reciprocity through endogamy, social sanctions, and reputational mechanisms that geographic proximity alone cannot sustain (see also [Munshi and Rosenzweig, 2016](#)). This evidence motivates my attention to caste-level results in the reduced-form analysis and to the role of social enforcement in the structural model.

The channels through which risk is shared informally have also received sustained attention. [Coate and Ravallion \(1993\)](#) and [Foster and Rosenzweig \(2001\)](#) study reciprocal transfers under limited commitment. [Udry \(1994\)](#) documents interpersonal lending as a key smoothing mechanism in northern Nigeria. [Ambrus et al. \(2014\)](#) shows that the topology of social networks determines the extent of insurance, and [Jackson et al. \(2012\)](#) formalises how network structure sustains cooperation through favour exchange. [Attanasio et al. \(2012\)](#) links risk preferences to network formation in Colombian villages. My paper contributes to this strand by proposing a specific mechanism: position-dependent enforcement through which social structure affects insurance outcomes.

## 2.2 Limited commitment and endogenous groups

The theoretical framework for my structural analysis builds on the dynamic limited commitment model of [Ligon et al. \(2002\)](#) (extending [Ligon et al., 2000](#); [Coate and Ravallion, 1993](#)). In this framework, a risk-sharing arrangement is sustained only if every household prefers to remain in the group rather than revert to autarky in every period and state. When a household's participation constraint binds, the planner adjusts Pareto weights to keep the household indifferent between staying and leaving. The model predicts partial insurance; consumption co-moves with income only when participation constraints bind and explains why risk sharing is incomplete even in communities with repeated interaction and public information.

A key limitation of the standard limited commitment model, identified by [Genicot and Ray \(2003\)](#), is that it assumes only individual deviations: a household either stays in the group or leaves alone. When coalitional deviations are allowed and a subgroup of households can jointly defect and form their own insurance arrangement, [Genicot and Ray \(2003\)](#) show that stable groups must be bounded in size. Intuitively, large groups offer better insurance (more diversification), but they also offer richer deviation possibilities: any sufficiently attractive subcoalition can break away. The tension between these forces produces an endogenous upper bound on group size.

[Bold and Broer \(2021\)](#) bring these insights to the ICRISAT data with a carefully calibrated structural model. Using a two-agent (one-vs-rest) representation of the [Ligon et al. \(2002\)](#) framework with coalitional deviations following [Genicot and Ray \(2003\)](#), they estimate the model by simulated method of moments on four consumption and income moments from the three ICRISAT villages. Their main findings are threefold. First, the standard limited commitment model with individual deviations produces counterfactually large asymmetry in consumption responses in village-sized groups: the model over-insures in bad states and under-insures in good states. Second, allowing for coalitional deviations endogenously produces small stable groups of 4 households, which dramatically improves the match to the data by generating approximately symmetric insurance. Third, the coalitional deviation model is preferred by the data across all three villages, with substantial improvements in the SMM criterion. These results establish the importance of enforcement structure i.e. who can deviate, and what happens when they do for the quantitative properties of limited commitment models.

Several extensions enrich the limited commitment framework. [Ábrahám and Laczó \(2018\)](#) add a storage technology and show that access to savings tightens participation constraints, reducing the scope for mutual insurance. [Laczó \(2015\)](#) introduces preference heterogeneity within the limited commitment framework and finds that it matters quantitatively for risk-sharing predictions. [Broer et al. \(2017\)](#) show that combining private information with limited enforcement avoids the extreme asymmetry of pure limited commitment models. Despite this progress, the literature has not explored how the severity of enforcement, specifically, how the cost of deviation varies with a household's economic position shapes insurance outcomes.

## 2.3 Social comparisons and relative position

The role of relative concerns in economic decisions has deep roots in economics and sociology. [Luttmer \(2005\)](#) provides causal evidence that neighbours' earnings reduce own subjective well-being, establishing that relative position matters independently of absolute income. [Fafchamps and Shilpi \(2008\)](#) extends this to a developing-country setting, showing that relative consumption affects subjective welfare in rural Nepal even after controlling for own consumption levels. [Stark and Taylor \(1991\)](#) shows that relative deprivation within a reference group is a powerful predictor of migration decisions, independent of absolute income levels. [Linssen et al. \(2011\)](#) and [Pattath \(2020\)](#) document that relative position affects a wide range of household outcomes including consumption, subjective well-being, and emigration in developing countries. [Bellet and Colson-Sihra \(2025\)](#) provide evidence that inequality distorts the consumption needs of the poor, while [Arcand and Pattath \(2023\)](#) show that relative deprivation affects consumption behaviour in developing country contexts.

In the Indian context, social comparisons are deeply embedded in the institution of caste. [Banerjee et al. \(2013\)](#) documents that caste governs marriage markets, [Rao \(2001\)](#) shows that public celebrations function as arenas for status competition in rural India, and [Bloch et al. \(2004\)](#) shows that wedding expenditure in particular serves as conspicuous consumption signalling social status within caste networks. [Munshi and Rosenzweig \(2009\)](#) argue that caste networks persist precisely because they provide insurance value that cannot be replicated through market mechanisms, and that this insurance function depends on within-caste social enforcement. These findings suggest that relative position within a caste group, which is likely both the reference group for comparisons and risk sharing group for transfers, is likely to affect the terms and sustainability of informal insurance.

Direct evidence on social comparisons and risk sharing is scarce but suggestive. [Gortner and van der Weele \(2019\)](#) show that information about others' portfolios increases risk sharing in experimental asset markets, demonstrating that knowledge of relative position changes insurance behaviour. [Fafchamps \(2011\)](#) documents that informal risk sharing in village economies is sustained by social norms, with deviations from expected behaviour triggering powerful sanctions including exclusion from community resources and loss of social capital. [Kaur \(2019\)](#) provides striking evidence of asymmetric social norms in Indian village labour markets: wages rise readily in response to positive productivity shocks but exhibit strong downward rigidity during droughts, enforced through worker sanctions against employers who cut pay. This asymmetry, that norms that protect vulnerable agents more strongly is the mechanism my structural model captures through position-dependent enforcement, where low-income households face harsher sanctions for deviating from the insurance arrangement.

My paper contributes by embedding position-dependent enforcement into a structural model of village risk sharing. By doing so, I bridge the gap between the reduced-form evidence that relative position predicts consumption outcomes and the structural limited commitment literature that has treated outside options as position-independent.

### 3 Reduced-Form Evidence: Relative Position and Risk Sharing

I first establish the empirical relevance of relative position for risk-sharing outcomes. I derive two theoretical specifications that motivate including measures of relative position in the standard risk-sharing test, and then take them to the ICRISAT data.

#### 3.1 Theoretical Motivation

My goal is to show that, under either of two natural specifications of preferences that incorporate relative concerns, the standard Townsend test suffers from omitted variable bias. The key insight is that it is not relative deprivation *per se* that belongs on the right-hand side, but rather a household's position in the empirical cumulative distribution function of consumption.

##### 3.1.1 Approach 1: relative satisfaction as an addendum to the utility function

I begin by augmenting agent preferences to include relative satisfaction in the sense of Yitzhaki.<sup>1</sup> This means that preferences will be given by:

$$U_i(c_{it}(s_t)) + \omega_i \int_0^{c_{it}(s_t)} [1 - F(z)] dz,$$

where  $S(c_{it}(s_t)) = \int_0^{c_{it}(s_t)} [1 - F(z)] dz$  is the relative satisfaction of individual  $i$ , where  $F(z)$  is the *cdf* of consumption in the population. In a standard Townsend framework, the Lagrangian facing the social planner is therefore given by:

$$\Lambda = \sum_{i=1}^{i=N} \left[ \sum_{t=1}^{t=T} \sum_{s_t=1}^{s_t=S} (\lambda_i \delta_i^{t-1} p_t(s_t) [U_i(c_{it}(s_t)) + \omega_i S(c_{it}(s_t))] - \mu_t(s_t) [c_{it}(s_t) - x_{it}(s_t)]) \right].$$

The FOCs associated with this Lagrangian for agents  $i$  and  $j$  are given by:

$$\lambda_i \delta_i^{t-1} p_t(s_t) [U'_i(c_{it}(s_t)) + \omega_i S'(c_{it}(s_t))] = \mu_t(s_t), \quad (1)$$

which can be rewritten as:

$$\lambda_i \delta_i^{t-1} p_t(s_t) [U'_i(c_{it}(s_t)) + \omega_i [1 - F(c_{it}(s_t))]] = \mu_t(s_t). \quad (2)$$

Rearranging terms, this can be written as:

$$U'_i(c_{it}(s_t)) = \frac{\mu_t(s_t)}{\lambda_i \delta_i^{t-1} p_t(s_t)} - \omega_i [1 - F(c_{it}(s_t))]. \quad (3)$$

Imposing a CARA or CRRA functional form then yields two empirical specifications. On the one hand, one can put  $c_{it}$  or  $\ln c_{it}$  on the LHS and consider a linear approximation: performing the usual exclusion restriction test in which household income enters on the RHS, I then note

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<sup>1</sup>The social evaluation of the deprivation inherent in a person's not having X is an increasing function of the proportion of those who do have it in the reference groups relevant to the person

that the standard specification suffers from omitted variable bias in that  $1 - F(c_{it}(s_t))$  would be missing as a covariate. Note that  $1 - F(c_{it}(s_t))$  can be evaluated directly from the appropriate empirical cumulative density function. Note also that relative deprivation *per se* should NOT appear on the RHS, but rather  $1 - F(c_{it}(s_t))$ .

### 3.1.2 Approach 2: Relative satisfaction right from the start

Now suppose that preferences are given by:

$$\int_0^{U_i(c_{it}(s_t))} [1 - F(z)] dz,$$

where  $S(c_{it}(s_t)) = \int_0^{c_{it}(s_t)} [1 - F(z)] dz$  is the relative satisfaction (Yitzhaki) of individual  $i$ . In a standard Townsend framework, the Lagrangian facing the social planner is therefore given by:

$$\Lambda = \sum_{i=1}^{i=N} \left[ \sum_{t=1}^{t=T} \sum_{s_t=1}^{s_t=S} \left( \lambda_i \delta_i^{t-1} p_t(s_t) \left[ \int_0^{U_i(c_{it}(s_t))} [1 - F(z)] dz \right] - \mu_t(s_t) [c_{it}(s_t) - x_{it}(s_t)] \right) \right].$$

The FOCs associated with this Lagrangian for agents  $i$  and  $j$  are given by:

$$\lambda_i \delta_i^{t-1} p_t(s_t) U'_i(c_{it}(s_t)) [1 - F(U_i(c_{it}(s_t)))] = \mu_t(s_t). \quad (4)$$

which can be rewritten as:

$$U'_i(c_{it}(s_t)) = \frac{\mu_t(s_t)}{\lambda_i \delta_i^{t-1} p_t(s_t) [1 - F(U_i(c_{it}(s_t)))]}. \quad (5)$$

The same discussion as for the first approach then applies, with the added twist that the specification of preferences now affects the argument of the cumulative density. However, under a relatively mild strict monotonicity (of the utility function) assumption (and as long as I assume that preferences are “close enough”, so that  $U_i(.) \approx U(.)$ ), while  $c_{it}(s_t)$  and  $U_i(c_{it}(s_t))$  are different,  $F(U_i(c_{it}(s_t)))$  and  $F(c_{it}(s_t))$  should be identical, allowing one to recover the value from the observable empirical cumulative density function of consumption.

### 3.1.3 Which rank?

Both derivations above establish that the standard Townsend test suffers from omitted variable bias: a household’s position in the cumulative distribution function belongs on the right-hand side. The preference-based derivation delivers *consumption* rank  $RP^c = 1 - F(c_{it})$  as the theoretically motivated object. In the empirical analysis below, I present results using both consumption rank and an alternative measure, *income* rank  $RP^y = 1 - F_y(y_{it})$ , constructed from the within-group empirical CDF of real total income per capita.

The two measures differ in important ways. Consumption rank is directly motivated by the theoretical model above, but it suffers from a mechanical negative correlation with the dependent variable  $\Delta c_{it}$ : a household that experienced a positive consumption change is, by construction,



higher-ranked. Income rank eliminates this problem because it is constructed from a different variable than the dependent variable. Income and consumption rank are strongly monotonically related in the data: the Spearman correlation exceeds 0.77 in all three villages.

I show that the central finding, that rank belongs in the Townsend test, is robust to the choice of rank measure. Both consumption rank and income rank are powerful predictors of consumption dynamics, and both eliminate rejections of efficient risk sharing. I present income rank alongside consumption rank because it provides a direct bridge to the structural model developed in Section 4, where enforcement depends on a household’s income-state rank.

### 3.2 Data

I use the ICRISAT Village Level Studies for three villages, Aurepalle, Shirapur, and Kanzara (Townsend, 1994). The sample comprises 40 households per village surveyed between July 1975 and July 1984/1985, selected to represent four land-holding classes. I use monthly data on consumption, labour supply, wages, demographic variables, and nonlabour income.

Consumption is constructed from the Household Transaction Schedule and consists of milled grain, other food items (oil, animal products, fruits and vegetables), and nondurable goods (electricity, water, cooking fuels). Following Townsend (1994) and Mazzocco and Saini (2012), I divide household consumption by an age-gender equivalence weight. Nonlabour income which is the sum of gifts, dowries, pensions, theft, and profits, serves as the excluded variable in the full insurance test.

Table 1 reports summary statistics for both waves of the survey. Panel A describes the sample used for the structural estimation: adult-equivalent consumption and income for a balanced panel over the crop years 1976–1981, the convention adopted by Townsend (1994) and Bold and Broer (2021) in view of concerns about income measurement in the later years. That sample contains 34, 37, and 31 households in Aurepalle, Shirapur, and Kanzara respectively, so 102 households and 612 household-year observations in total. Panel B describes the VLS2 resurvey of the same three villages used in the out-of-sample exercise of Appendix G, which is wider in the cross-section and much shorter in the time dimension. Levels are not comparable across the two panels. Two features of Panel A are worth noting for what follows. Income is considerably more variable than consumption in every village, which is the elementary sign that some smoothing is taking place; and the dispersion of consumption across households within a village is substantial, which is the moment the structural model of Section 4 is asked to reproduce.

### 3.3 Empirical Specifications

I estimate the standard homogeneous-preference test:

$$\Delta\rho_t^i = \alpha_0 + \alpha_1 X_t + \alpha_2 \Delta\rho_t^a + \alpha_3 \Delta y_t^i, \quad (6)$$

**Table 1**

Summary Statistics: ICRISAT VLS1 and VLS2

|                                                                    |    |     |     | Adult  | Land | Consumption |       | Income |       | Std. dev.      |                |
|--------------------------------------------------------------------|----|-----|-----|--------|------|-------------|-------|--------|-------|----------------|----------------|
| Village                                                            | HH | $T$ | $N$ | equiv. | (ha) | Mean        | S.D.  | Mean   | S.D.  | $\Delta \ln c$ | $\Delta \ln y$ |
| <i>Panel A: VLS1, 1976–1981 (adult-equivalent, balanced panel)</i> |    |     |     |        |      |             |       |        |       |                |                |
| Aurepalle                                                          | 34 | 6   | 204 | 5.95   | 0.60 | 303         | 128   | 630    | 430   | 0.26           | 0.48           |
| Shirapur                                                           | 37 | 6   | 222 | 5.66   | 0.75 | 401         | 161   | 984    | 743   | 0.27           | 0.36           |
| Kanzara                                                            | 31 | 6   | 186 | 5.85   | 0.92 | 430         | 171   | 792    | 578   | 0.31           | 0.54           |
| <i>Panel B: VLS2, 2001–2004 (per capita, winsorised at 1%)</i>     |    |     |     |        |      |             |       |        |       |                |                |
| Aurepalle                                                          | 94 | 4   | 318 | —      | —    | 7,929       | 5,410 | 8,107  | 5,552 | 0.55           | 0.80           |
| Shirapur                                                           | 87 | 4   | 293 | —      | —    | 7,793       | 5,382 | 9,239  | 8,857 | 0.64           | 1.19           |
| Kanzara                                                            | 51 | 4   | 178 | —      | —    | 6,846       | 3,693 | 7,723  | 8,101 | 0.40           | 1.04           |

*Notes:* Panel A uses the estimation sample underlying the data moments in Table 6: adult-equivalent consumption and income, balanced panel, crop years 1976–1981, each village separately, following the conventions in Townsend (1994) and Bold and Broer (2021). Panel B uses the VLS2 replication data of Morten (2019), restricted to the same three villages, per capita and winsorised at the 1st and 99th percentiles by village. Consumption and income are in the deflated units of each wave and are *not* comparable across panels. The final two columns report standard deviations of first differences after removing the village-year mean, the transformation used to construct  $m_1$  and  $m_2$ .

where  $\Delta \rho_t^i$ ,  $\Delta \rho_t^a$ , and  $\Delta y_t^i$  are the first differences in household expenditure, village/caste expenditure, and household income, respectively, and  $X_t$  is a vector of controls. The two approaches yield modified specifications:

$$\Delta \rho_t^i = \alpha_0 + \alpha_1 X_t + \alpha_2 \Delta \rho_t^a + \alpha_3 \Delta y_t^i + \zeta RP_{vc}, \quad (7)$$

$$\Delta \rho_t^i = \alpha_0 + \alpha_1 X_t + \alpha_2 \Delta \rho_t^a + \alpha_3 \Delta y_t^i + \zeta RP_{vc} + \gamma RP_{vc} \times \Delta y_t^i, \quad (8)$$

where  $RP_{vc}$  is the household’s relative position within the village or caste. I implement two measures:

$$RP_{vc,t}^c = 1 - F_c(c_{it}), \quad (9)$$

$$RP_{vc,t}^y = 1 - F_y(y_{it}), \quad (10)$$

where  $F_c(\cdot)$  and  $F_y(\cdot)$  are the within-group (village or caste), within-year empirical CDFs of real total expenditure per adult unit and real total income per capita, respectively. A household with the highest consumption (or income) in its group has  $RP \approx 0$ ; the lowest has  $RP \approx 1$ . Consumption rank is the object directly motivated by the theoretical model; income rank provides a mechanical-correlation-free alternative and a direct bridge to the structural model (Section 4).

### 3.4 Results

The central finding is that rank is a predictor of consumption dynamics, and controlling for it substantially reduces or eliminates rejections of efficient risk sharing. Tables 2 and 3 report the results using consumption rank and income rank, respectively.

**Consumption rank (Table 2).** Consumption rank  $RP^c$  is significant at the 1% level across all three villages and all 9 castes, in both Approach 1 and Approach 2. The coefficients are large in magnitude:  $-82$  (Aurepalle),  $-125$  (Shirapur),  $-149$  (Kanzara). Under Approach 2, the income coefficient becomes insignificant in 2 of 3 villages and 7 of 9 castes, indicating that controlling for consumption rank largely eliminates rejections of efficient risk sharing.

However, because  $RP^c = 1 - F(c_{it})$  is a function of contemporaneous consumption and the dependent variable is  $\Delta c_{it}$ , a mechanical negative correlation arises: households with high current consumption tend to have experienced positive  $\Delta c$  and therefore have high rank (low  $RP^c$ ). Thus the large negative coefficient on  $RP^c$  partly reflects this mechanical relationship and cannot be given a causal interpretation. I treat the consumption-rank regressions as descriptive evidence that motivates both the income-rank specification and the structural investigation.

**Income rank (Table 3).** Income rank  $RP^y$  confirms the finding while eliminating the mechanical correlation concern. Because  $RP^y$  is constructed from the income distribution rather than the consumption distribution, there is no mechanical relationship with  $\Delta c_{it}$ . Income rank is significant at the 1% level across all three villages, with negative coefficients:  $-54$  (Aurepalle),  $-85$  (Shirapur),  $-93$  (Kanzara). Adding income rank roughly doubles the  $R^2$ , from 0.14 to 0.29 (Aurepalle), 0.11 to 0.29 (Shirapur), and 0.15 to 0.33 (Kanzara). At the caste level, income rank is significant at the 1% level in all 9 castes under asymptotic cluster-robust inference.

Because the caste-level regressions have as few as 5–15 household clusters, asymptotic cluster-robust standard errors are downward-biased and over-reject (Cameron et al., 2008). Table 3 therefore also reports wild cluster bootstrap  $p$ -values (Webb six-point weights, 9,999 replications, imposing the null), which remain valid with a small number of clusters (Canay et al., 2021; Webb, 2023). The rank coefficient is robust to this correction: the wild bootstrap  $p$ -value for  $RP^y$  is below 0.02 in every village and every caste. The bootstrap does, however, discipline the smaller caste-level income coefficients, as discussed below.

The coefficients on income rank are smaller in magnitude than those on consumption rank ( $-54$  to  $-93$  vs.  $-82$  to  $-149$ ), as expected: the consumption-rank coefficients are inflated by the mechanical correlation. The fact that income rank remains highly significant with large coefficients even after removing the mechanical component could indicate that the finding reflects a genuine empirical pattern, not an artifact.

Under Approach 2 with income rank, the coefficient on  $\Delta y^{nl}$  becomes insignificant in all three villages ( $p = 0.20, 0.23, 0.27$ ), eliminating rejections of efficient risk sharing at the village level. The interaction term  $RP^y \times \Delta y^{nl}$  is itself not significant ( $p > 0.30$  in all villages), indicating that the position effect operates through the *level* of consumption dynamics rather than through differential sensitivity to income shocks.

At the caste level, one apparent exception deserves comment. Under asymptotic cluster-robust inference, the Mahar caste in Kanzara is the only group in which the income coefficient remains significant under Approach 2 (0.003, nominal  $p < 0.01$ ). Anomalously, its standard error *falls* relative to Approach 1 (from 0.005 to 0.001) when the interaction is added, the opposite of

every other row in the table. Both features are artifacts of few-cluster asymptotics: the Mahar subsample contains only 6 households, the heteroskedasticity-robust standard error on the same coefficient is twenty times larger (0.024), and dropping a single household moves the coefficient from 0.003 to 0.112. The wild cluster bootstrap  $p$ -value is 0.12, so the Mahar rejection does not survive inference that is valid with few clusters. Under wild bootstrap inference, controlling for income rank eliminates rejections of efficient risk sharing in *all* three villages and *all* nine castes. Appendix E further confirms the pattern using lagged income rank  $RP_{i,t-1}^y$ , which produces positive, highly significant coefficients (+18 to +21), indicating that households with higher past income experience larger subsequent consumption growth.

The income-rank results, being free of mechanical correlation, provide the cleaner identification and the more direct bridge to the structural model, which conditions enforcement on income-state rank (Section 4). Table 4 summarises rejection patterns.

These results are consistent with caste as the relevant risk-sharing unit, aligning with Grimard (1997), Mazzocco and Saini (2012), and Munshi (2019). The findings are broadly robust to alternative consumption categories and testing specifications (Appendix A), to GMM estimation (Appendix B), to instrumenting consumption rank (Appendix D), and to out-of-sample replication on the ICRISAT VLS2 data (Appendix G).<sup>2</sup>

### 3.5 From Reduced Form to Structure

While the reduced-form evidence shows that a household’s rank is a predictor of consumption dynamics, and that controlling for it eliminates rejections of efficient risk sharing, the regressions cannot distinguish between competing mechanisms. Relative position may matter because households have preferences over their relative consumption (as in the theoretical motivation above), or because communities enforce insurance arrangements through sanctions that depend on a household’s economic standing, or through some combination of both channels. To disentangle these mechanisms and quantify their importance, I turn to a structural model that traces the effects of position-dependent forces on equilibrium allocations, group formation, and the joint distribution of consumption and income.

The natural starting point is Bold and Broer’s (2021) limited commitment model with coalitional deviations. Their framework is uniquely suited to my question. It is the only structural model of village risk sharing that simultaneously addresses endogenous group formation, coalitional stability, and the quantitative moments of the ICRISAT consumption data that I use. Its central mechanism operates through participation constraints and outside options, since whether a household stays in the insurance group depends on the value of autarky relative to continued membership, which makes the model a natural vehicle for studying how enforcement intensity, i.e. the cost of deviation, varies with economic position. And Bold and Broer (2021)’s key finding that coalitional deviations produce small stable groups (4 households) with approximately symmetric insurance provides a well-identified baseline against which to measure the

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<sup>2</sup>Using the ICRISAT VLS2 (2001–2004) from Morten (2019), relative position is significant at the 1% level in all three villages, and the income coefficient under Approach 2 is insignificant in all villages and castes.

**Table 2**  
Risk Sharing and Consumption Rank

| Village/Caste | Mazzocco and Saini (2012) | Approach 1                         |                        | Approach 2                         |                        | N    |
|---------------|---------------------------|------------------------------------|------------------------|------------------------------------|------------------------|------|
|               |                           | Coefficient on<br>Nonlabour income | Relative Position      | Coefficient on<br>Nonlabour income | Relative Position      |      |
| Aurepalle     | 0.017***<br>(0.004)       | 0.012***<br>(0.003)                | −82.436***<br>(1.982)  | 0.006<br>(0.008)                   | −82.397***<br>(1.983)  | 3635 |
| Gowda         | 0.030***<br>(0.012)       | 0.015*<br>(0.009)                  | −105.092***<br>(4.044) | 0.023<br>(0.016)                   | −105.297***<br>(4.059) | 944  |
| Kurba         | 0.025<br>(0.021)          | 0.010<br>(0.015)                   | −69.050***<br>(2.817)  | 0.003<br>(0.022)                   | −68.977***<br>(2.824)  | 595  |
| Madiga        | −0.084<br>(0.147)         | −0.060<br>(0.122)                  | −60.865***<br>(3.724)  | −0.004<br>(0.204)                  | −60.926***<br>(3.731)  | 589  |
| Reddi         | 0.006*<br>(0.003)         | 0.008***<br>(0.003)                | −38.744***<br>(3.063)  | 0.007<br>(0.005)                   | −38.727***<br>(3.067)  | 595  |
| Kanzara       | 0.009*<br>(0.005)         | 0.007**<br>(0.003)                 | −149.095***<br>(2.795) | 0.010<br>(0.007)                   | −149.091***<br>(2.796) | 3677 |
| Mahar         | 0.008<br>(0.012)          | 0.012<br>(0.010)                   | −134.636***<br>(7.256) | 0.031**<br>(0.014)                 | −133.921***<br>(7.251) | 611  |
| Mali          | 0.031**<br>(0.013)        | 0.017*<br>(0.010)                  | −129.355***<br>(4.105) | 0.026<br>(0.019)                   | −129.436***<br>(4.108) | 1376 |
| Maratha       | 0.005<br>(0.006)          | 0.002<br>(0.005)                   | −136.325***<br>(5.873) | −0.006<br>(0.008)                  | −136.514***<br>(5.873) | 738  |
| Shirapur      | 0.009***<br>(0.003)       | 0.005**<br>(0.002)                 | −125.017***<br>(2.866) | 0.011***<br>(0.004)                | −125.047***<br>(2.865) | 3243 |
| Dhangar       | 0.010**<br>(0.004)        | 0.007**<br>(0.003)                 | −111.532***<br>(4.815) | 0.008*<br>(0.004)                  | −111.535***<br>(4.818) | 818  |
| Maratha       | 0.004<br>(0.003)          | 0.000<br>(0.003)                   | −126.660***<br>(4.502) | 0.001<br>(0.007)                   | −126.662***<br>(4.503) | 1576 |

*Notes:*  $RP^c = 1 - F(c_{it})$  is consumption rank. Standard errors in parentheses, clustered at the household level. \*p<0.1; \*\*p<0.05; \*\*\*p<0.01. Village-level rows include all households; caste-level rows restrict to the indicated caste. Castes with fewer than 5 households are excluded (Mala and Brahmin in Aurepalle, Mahar in Shirapur, Gosavi in Kanzara). Caste-level rows have between 5 and 15 household clusters, so asymptotic cluster-robust standard errors over-reject (Cameron et al., 2008) and caste-level significance markers should be read with caution; Table 3 reports wild cluster bootstrap  $p$ -values for the income-rank specifications. Important caveat: because  $RP^c$  is a function of contemporaneous consumption and the dependent variable is  $\Delta c_{it}$ , the coefficient on  $RP^c$  reflects a mechanical negative correlation and cannot be given a causal interpretation (see Section 3).

**Table 3**  
Risk Sharing and Income Rank

| Village/Caste | Mazzocco and Saini (2012)      | Approach 1                         |                                                         | Approach 2                                              |                                   | N    | H  |
|---------------|--------------------------------|------------------------------------|---------------------------------------------------------|---------------------------------------------------------|-----------------------------------|------|----|
|               |                                | Coefficient on<br>Nonlabour income | Coefficient on<br>Nonlabour income    Relative Position | Coefficient on<br>Nonlabour income    Relative Position |                                   |      |    |
| Aurepalle     | 0.017***<br>(0.003)<br>[<0.01] | 0.014***<br>(0.003)<br>[0.01]      | −53.693***<br>(6.416)<br>[<0.01]                        | 0.009<br>(0.007)<br>[0.28]                              | −53.707***<br>(6.414)<br>[0.34]   | 3635 | 31 |
| Gowda         | 0.030***<br>(0.009)<br>[<0.01] | 0.015**<br>(0.007)<br>[0.17]       | −65.103***<br>(9.219)<br>[<0.01]                        | 0.029<br>(0.022)<br>[0.28]                              | −65.346***<br>(9.134)<br>[0.64]   | 944  | 8  |
| Kurba         | 0.025<br>(0.034)<br>[0.48]     | 0.021<br>(0.030)<br>[0.50]         | −35.807***<br>(2.839)<br>[0.01]                         | 0.012<br>(0.041)<br>[0.83]                              | −35.798***<br>(2.810)<br>[0.70]   | 595  | 5  |
| Madiga        | −0.084<br>(0.220)<br>[0.86]    | −0.028<br>(0.109)<br>[0.72]        | −50.256***<br>(10.092)<br>[0.02]                        | −0.108<br>(0.271)<br>[0.80]                             | −50.201***<br>(10.188)<br>[0.70]  | 589  | 5  |
| Reddi         | 0.006<br>(0.005)<br>[0.37]     | 0.007<br>(0.004)<br>[0.10]         | −15.795***<br>(1.444)<br>[0.01]                         | 0.006<br>(0.007)<br>[0.45]                              | −15.798***<br>(1.468)<br>[0.96]   | 595  | 5  |
| Kanzara       | 0.009***<br>(0.003)<br>[0.01]  | 0.010***<br>(0.003)<br>[<0.01]     | −92.663***<br>(10.860)<br>[<0.01]                       | 0.008<br>(0.008)<br>[0.36]                              | −92.684***<br>(10.877)<br>[0.80]  | 3677 | 35 |
| Mahar         | 0.008***<br>(0.003)<br>[0.30]  | 0.009*<br>(0.005)<br>[0.28]        | −100.721***<br>(10.498)<br>[0.01]                       | 0.003***<br>(0.001)<br>[0.12]                           | −101.224***<br>(10.537)<br>[0.20] | 611  | 6  |
| Mali          | 0.031**<br>(0.013)<br>[0.08]   | 0.024*<br>(0.014)<br>[0.20]        | −90.600***<br>(11.599)<br>[<0.01]                       | −0.006<br>(0.025)<br>[0.82]                             | −90.536***<br>(11.551)<br>[0.13]  | 1376 | 13 |
| Maratha       | 0.005<br>(0.005)<br>[0.45]     | 0.004<br>(0.005)<br>[0.48]         | −45.079***<br>(9.535)<br>[<0.01]                        | −0.007<br>(0.010)<br>[0.65]                             | −45.473***<br>(9.485)<br>[0.39]   | 738  | 7  |
| Shirapur      | 0.009**<br>(0.004)<br>[0.01]   | 0.006*<br>(0.003)<br>[0.01]        | −85.181***<br>(8.203)<br>[<0.01]                        | 0.006<br>(0.005)<br>[0.30]                              | −85.179***<br>(8.207)<br>[0.89]   | 3243 | 31 |
| Dhangar       | 0.010***<br>(0.002)<br>[0.01]  | 0.009***<br>(0.002)<br>[0.09]      | −63.731***<br>(5.609)<br>[<0.01]                        | 0.005<br>(0.005)<br>[0.43]                              | −63.406***<br>(5.413)<br>[0.69]   | 818  | 8  |
| Maratha       | 0.004*<br>(0.002)<br>[0.14]    | 0.001<br>(0.002)<br>[0.54]         | −81.893***<br>(11.822)<br>[<0.01]                       | −0.002<br>(0.008)<br>[0.85]                             | −81.964***<br>(11.787)<br>[0.79]  | 1576 | 15 |

*Notes:*  $RP^y = 1 - F_y(y_{it})$  is the household's income rank: one minus the within-group, within-year ECDF of real total income per capita. Village-level rows include all households; caste-level rows restrict to the indicated caste. Castes with fewer than 5 households are excluded (Mala and Brahmin in Aurepalle, Mahar in Shirapur, Gosavi in Kanzara).  $N$  is household-period observations;  $H$  is the number of households, i.e. clusters. Standard errors in parentheses, clustered at the household level; significance stars are based on these asymptotic cluster-robust standard errors. Because the number of clusters is small, especially at the caste level ( $H \leq 15$ ), asymptotic cluster-robust inference over-rejects (Cameron et al., 2008); square brackets therefore report wild cluster bootstrap  $p$ -values (Webb six-point weights, 9,999 replications, null imposed), which are reliable with few clusters (Canay et al., 2021). \* $p < 0.1$ ; \*\* $p < 0.05$ ; \*\*\* $p < 0.01$ .

**Table 4**

Rejection of Efficient Risk Sharing

|                                                                              | MS  | Approach 1 | Approach 2 |
|------------------------------------------------------------------------------|-----|------------|------------|
| <i>How often is <math>\Delta y^{nl}</math> significant at the 10% level?</i> |     |            |            |
| <i>Panel A: Consumption rank (<math>RP^c</math>)</i>                         |     |            |            |
| Village                                                                      | 3/3 | 3/3        | 1/3        |
| Castes                                                                       | 4/9 | 4/9        | 2/9        |
| <i>Panel B: Income rank (<math>RP^y</math>)</i>                              |     |            |            |
| Village                                                                      | 3/3 | 3/3        | 0/3        |
| Castes                                                                       | 5/9 | 4/9        | 1/9        |

*Notes:* Panel A uses consumption rank  $RP^c = 1 - F(c_{it})$  from Table 2; Panel B uses income rank  $RP^y = 1 - F_y(y_{it})$  from Table 3. Both panels use 9 castes with  $\geq 5$  households (see table notes). Counts are based on asymptotic cluster-robust standard errors. Under wild cluster bootstrap inference (Table 3), which is valid with few clusters, the Panel B counts become: Village 3/3 (MS), 3/3 (Approach 1), 0/3 (Approach 2); Castes 3/9 (MS), 1/9 (Approach 1), 0/9 (Approach 2); in particular, the single caste-level Approach 2 rejection under asymptotic inference (Mahar, Kanzara) has a wild bootstrap  $p$ -value of 0.12. Under Approach 2, rank is significant at the 1% level in all villages and all 9 castes in both panels, and its wild bootstrap  $p$ -value is below 0.02 in every row. The interaction  $RP^y \times \Delta y^{nl}$  is insignificant in all income-rank specifications. Appendix A reports additional rejection counts across consumption categories.

effect of position-dependent enforcement.

I choose to model the enforcement channel, position-dependent sanctions on outside options, rather than the preference channel. Doing so is computationally tractable within the [Bold and Broer \(2021\)](#) framework, because preferences remain standard CRRA and only the autarky values are modified, preserving the Borch-rule allocation conditional on binding patterns. Position-dependent enforcement also has independent micro-foundations in the anthropological and development economics literatures on Indian village sanctions: [Fafchamps \(2011\)](#) documents that community enforcement of informal arrangements depends on social capital, which the poor possess less of, and [Kaur \(2019\)](#) shows that social norms in Indian villages asymmetrically protect vulnerable agents. Beyond tractability and micro-foundations, the enforcement channel generates testable predictions about cross-sectional consumption dispersion, since households with different ranks face different outside options, producing heterogeneous consumption levels in equilibrium that the preference channel, with its standard CRRA sharing rule, does not.

## 4 Structural Model: Limited Commitment with Position-Dependent Enforcement

[Bold and Broer \(2021\)](#) extend the [Ligon et al. \(2002\)](#) limited commitment framework with coalitional deviations following [Genicot and Ray \(2003\)](#). I introduce a single modification: the

social sanctions imposed on deviators are position-dependent.

#### 4.1 Environment

Consider a village economy with a finite set of households  $\mathcal{N} = \{1, \dots, N\}$ . Each period  $t$ , household  $i$  receives an idiosyncratic endowment  $y_{it} \in \mathcal{Y} = \{y^1, \dots, y^K\}$ . Endowments follow a finite-state Markov process with common transition matrix  $Q$ :

$$\Pr\{y_{i,t+1} = y' \mid y_{it} = y\} = Q(y', y) \quad \text{for all } i.$$

Conditional on own lagged income, realisations are independent across households. All incomes, transfers, and coalition membership are publicly observed each period.

#### 4.2 Preferences

Preferences are standard CRRA:

$$u(c) = \begin{cases} \frac{c^{1-\rho}}{1-\rho}, & \rho \neq 1, \\ \ln c, & \rho = 1, \end{cases} \quad \rho > 0. \quad (11)$$

In estimation, I fix  $\rho = 1$  (log utility) throughout, so the model has two free parameters: the discount factor  $\delta$  and the enforcement intensity  $\omega$ . There is no direct preference over relative position. The role of income-state rank enters exclusively through the enforcement mechanism described below.

#### 4.3 Two-Agent Representation

I use a two-agent (one-vs-rest) representation of a coalition of size  $n$ : agent 1 is a single household, and agent 2 represents a typical member of the remaining  $n - 1$  households. Per-household quantities  $(y_{2t}, c_{2t})$  apply to each of the  $n - 1$  remaining members. This aggregation is exact under ex-ante symmetry: all households draw from the same income process, so the  $n - 1$  non-focal agents are interchangeable and subcoalitions are characterised by size, not composition (Bold and Broer, 2021; Genicot and Ray, 2003). Aggregate resources in a coalition of size  $n$  are:

$$Y^{(n)} = y_1 + (n - 1)y_2. \quad (12)$$

#### 4.4 Risk-Sharing Arrangement

Within a coalition of size  $n$ , a limited-commitment planner chooses consumption allocations  $\{c_1, c_2\}$  to maximise the weighted sum of discounted utilities, subject to feasibility and participation constraints.



**Feasibility.** Each period,

$$c_1 + (n - 1)c_2 = Y^{(n)}. \quad (13)$$

**Recursive problem.** Let  $V^{(n)}(s; \lambda)$  be the continuation value for a coalition of size  $n$  in state  $s = (y_1, y_2)$  with Pareto weight  $\lambda \in (0, 1)$  on agent 1. The Bellman equation is:

$$V^{(n)}(s; \lambda) = \max_{c_1, c_2, \lambda'(s')} \left\{ \lambda u(c_1) + (1 - \lambda) u(c_2) + \delta \mathbb{E} \left[ V^{(n)}(s'; \lambda'(s')) \mid s \right] \right\} \quad (14)$$

subject to (13) and the participation constraints below.

## 4.5 Position-Dependent Enforcement

When a household deviates from the insurance arrangement, the community imposes social sanctions. The key innovation of my model is that these sanctions are *position-dependent*: a household's punishment for defection depends on its position in the within-coalition income distribution.

**Income-state rank.** For a household with income  $y_{it} \in \mathcal{Y} = \{y^1, \dots, y^K\}$  (ordered  $y^1 < \dots < y^K$ ), define its income-state rank within the coalition as:

$$r(y_{it}) = \frac{\text{rank of } y_{it} \text{ in } \mathcal{Y}}{K} \in \left\{ \frac{1}{K}, \frac{2}{K}, \dots, 1 \right\}. \quad (15)$$

Note that  $r(\cdot)$  is a rank over the *fixed income support*  $\mathcal{Y}$ , not the realised within-coalition consumption rank. This is a structural requirement, not merely a modelling convenience: in the planner's problem, consumption is the *endogenous* allocation being solved for, while income states are *exogenous* primitives. Conditioning enforcement on consumption rank would create a circularity (sanctions would affect allocations, which determine consumption rank, which in turn determines sanctions), requiring a fixed-point solution of a fundamentally different character.<sup>3</sup> With  $K = 3$  states, income-state rank is more precisely an income-state index. Under partial insurance, income-state rank and realised consumption rank are strongly correlated but not identical; under partial insurance, the Spearman correlation between income rank and consumption rank exceeds 0.77 in all three villages.

**Sanctions function.** The fraction of income lost to sanctions upon deviation is:

$$\sigma(y_{it}; \omega) = \omega (1 - r(y_{it})), \quad \omega \in [0, 1]. \quad (16)$$

---

<sup>3</sup>In the canonical limited commitment models (Kocherlakota, 1996; Ligon et al., 2002; Kehoe and Levine, 1993), the outside option is always the value of autarky, which depends on the exogenous income process by construction. Marcet and Marimon (2019) note that endogenous outside options “require solving a fixed-point problem.” Even making the outside option depend on endogenous *capital* rather than consumption creates significant complications (Kehoe and Perri, 2002; Abraham and Carceles-Poveda, 2006).

The parameter  $\omega$  governs the intensity of income-state-dependent enforcement. With  $K = 3$  income states, rank takes values  $\{1/3, 2/3, 1\}$ , so the sanctions function distinguishes three levels of enforcement. This is coarser than the continuous income rank used in the reduced-form analysis ( $RP^y = 1 - F_y(y_{it})$ ), but matches the discretisation required by the [Bold and Broer \(2021\)](#) computational framework. When  $\omega = 0$ , there are no sanctions and the model reduces to Bold and Broer’s baseline. When  $\omega > 0$ :

- The highest-income household ( $r = 1$ ) faces zero sanctions.
- The lowest-income household ( $r = 1/K$ ) faces sanctions  $\omega(1 - 1/K)$ .
- Intermediate households face sanctions proportional to their distance from the top.

**Autarky consumption.** The consumption available to a deviating household in income state  $y_{it}$  is:

$$c_{it}^{\text{aut}} = y_{it} (1 - \sigma(y_{it}; \omega)) = y_{it} (1 - \omega(1 - r(y_{it}))). \quad (17)$$

Sanctions are permanent: a household that leaves the coalition consumes  $c^{\text{aut}}$  in every subsequent period, so the continuation value of autarky is  $W^{\text{aut}}(s) = \sum_{t=0}^{\infty} \delta^t \mathbb{E}[u(c_t^{\text{aut}}) \mid s_0 = s]$ , computed as  $(I - \delta \Pi)^{-1} u(c^{\text{aut}})$  where  $\Pi$  is the income transition matrix.

**Economic interpretation.** The sanctions function does not assume that communities deliberately punish the poor more harshly. Rather, it captures the fact that the same act i.e. exclusion from the insurance group imposes asymmetric costs depending on a household’s economic position. Three features of village enforcement, documented in the anthropological and development economics literatures, motivate this formulation:

- **Asymmetric vulnerability to exclusion.** [Fafchamps \(2011\)](#) argues that enforcement of informal risk-sharing arrangements operates through exclusion from networks, loss of access to common resources, and reputational sanctions. Wealthier households can substitute away from these community-provided goods through formal credit, asset sales, or migration. Poorer households cannot: community membership provides access to mutual aid, common grazing land, informal credit, and emergency assistance that collectively represent a large share of their effective income. The same sanction therefore has a position-dependent bite.
- **Bargaining power and outside options.** High-income households can credibly threaten to leave because autarky is tolerable for them. The community has less leverage over agents whose outside option is already attractive. Low-income households, by contrast, depend on the group for survival, giving the community substantial enforcement power over them.
- **Coalitional deviations and sub-group insurance.** A key feature of the [Bold and Broer \(2021\)](#) framework is that the relevant outside option is not pure autarky but the value of forming a *smaller sub-coalition*. Even a poor household can deviate as part of

a sub-group and obtain some insurance. Position-dependent sanctions must therefore be strong enough to offset the attractiveness of these sub-coalitional deviations. This explains why the estimated  $\omega$  is near its upper bound: without substantial sanctions, low-income households can credibly threaten to form breakaway sub-groups, unravelling the insurance arrangement.

#### 4.6 Participation Constraints

Under individual deviations (ID), a household can unilaterally leave the coalition. The deviator becomes a singleton and the remaining  $n - 1$  households form their own coalition. The participation constraints, written in the recursive (state-contingent) form consistent with (14), are:

$$u(c_1) + \delta \mathbb{E} \left[ W_1^{(n)}(s'; \lambda') \mid s \right] \geq u(c_1^{\text{aut}}) + \delta \mathbb{E} \left[ W_1^{\text{out}}(s') \mid s \right], \quad (18)$$

$$u(c_2) + \delta \mathbb{E} \left[ W_2^{(n)}(s'; \lambda') \mid s \right] \geq u(c_2^{\text{aut}}) + \delta \mathbb{E} \left[ W_2^{\text{out}}(s') \mid s \right], \quad (19)$$

Here  $W_i^{(n)}(s'; \lambda')$  is agent  $i$ 's continuation value inside the coalition, obtained from the Bellman equation (14):  $W_1^{(n)}(s; \lambda) \equiv V^{(n)}(s; \lambda)$ , and  $W_2^{(n)}$  is defined symmetrically with weight  $(1 - \lambda)$ . The outside-option value  $W_i^{\text{out}}(s')$  is the value of the optimal coalitionally-stable arrangement available to the deviator after leaving the original coalition (computed recursively at smaller coalition sizes); the base case is  $n = 1$  (autarky), where  $W_i^{\text{aut}}(s) = (I - \delta \Pi)^{-1} u(c^{\text{aut}})$  as defined above.

Note that position-dependent sanctions enter the participation constraints through  $c^{\text{aut}}$ , equation (17). For low-income households, sanctions reduce  $c^{\text{aut}}$ , relaxing the participation constraint (making deviation less attractive). For high-income households,  $\sigma \approx 0$  and the constraint is essentially unchanged from the baseline.

**Intraperiod FOCs.** Let  $\mu$  be the multiplier on (13), and  $\phi_1, \phi_2 \geq 0$  the multipliers on (18)–(19). The intraperiod first-order conditions are:

$$(\lambda + \phi_1) u'(c_1) = \mu, \quad (20)$$

$$((1 - \lambda) + \phi_2) u'(c_2) = (n - 1) \mu. \quad (21)$$

The  $(n - 1)$  factor on the right-hand side of (21) reflects the shadow cost of feasibility: raising per-capita consumption  $c_2$  by one unit requires  $(n - 1)$  units of aggregate resources. Dividing (20) by (21) eliminates  $\mu$  and yields the standard CRRA sharing rule,

$$\frac{u'(c_1)}{u'(c_2)} = \frac{(n - 1)((1 - \lambda) + \phi_2)}{\lambda + \phi_1}, \quad (22)$$

which determines the consumption ratio as a function of the effective Pareto weights. The effect of  $\omega$  enters only through which constraints bind and how often, and therefore through  $\phi_1, \phi_2$ ; it does not enter through the allocation rule itself.

## 4.7 Coalitional Deviations and Endogenous Group Size

Following [Genicot and Ray \(2003\)](#) and [Bold and Broer \(2021\)](#), I require that the insurance arrangement be robust not just to individual deviations, but to deviations by any subcoalition  $\mathcal{D} \subset \mathcal{C}$ . The deviating subcoalition  $\mathcal{D}$  forms its own insurance arrangement and must itself be robust to further subcoalitional deviations. This recursive stability requirement bounds the maximal sustainable group size from above.

Formally, for every subcoalition  $\mathcal{D} \subset \mathcal{C}$  and every household  $i \in \mathcal{D}$ :

$$u(c_i) + \delta \mathbb{E} [W_i^{\text{in}}(s'; \mathcal{C}) \mid s] \geq u(c_i^{\text{dev}}(\mathcal{D})) + \delta \mathbb{E} [W_i^{\text{out}}(s'; \mathcal{D}) \mid s], \quad (23)$$

where  $c_i^{\text{dev}}(\mathcal{D})$  is the consumption available to household  $i$  upon deviating as part of coalition  $\mathcal{D}$ , with sanctions applied based on  $i$ 's income-state rank. Sanctions are imposed by the *remaining* members of the original coalition on the deviating subcoalition: the community punishes defection by restricting access to resources, social capital, and informal credit. Within the deviating subcoalition  $\mathcal{D}$ , members form their own insurance arrangement (which must itself be coalitionally stable), but their outside options upon entering this new arrangement reflect the sanctions incurred by leaving the original group. The maximum coalition size  $n^*$  such that (23) holds for all subcoalitions is the endogenous group size.

## 5 Estimation

### 5.1 Income Process

The individual income process is estimated from the ICRISAT data as an AR(1) in log income, with parameters summarised in Table 5. The near-zero persistence across all three villages is consistent with [Bold and Broer \(2021\)](#): the income process is well approximated by an i.i.d. process, which I discretise into three states using the [Rouwenhorst \(1995\)](#) method. The relatively high standard deviations of the innovation ( $\sigma_\varepsilon \approx 0.50\text{--}0.67$ ) confirm that these are risky environments with substantial idiosyncratic income variation, the kind of setting where informal insurance mechanisms should be most valuable ([Townsend, 1995](#)).

**Table 5**  
Estimated Income Processes

|                                | Aurepalle | Shirapur | Kanzara |
|--------------------------------|-----------|----------|---------|
| Persistence ( $\hat{\rho}_1$ ) | 0.039     | 0.003    | −0.021  |
| Var(log $y$ )                  | 0.199     | 0.079    | 0.121   |
| Households                     | 40        | 40       | 40      |
| Period                         | 1975–84   | 1975–84  | 1975–85 |

*Notes:* Income process estimated from ICRISAT VLS data following ([Bold and Broer, 2021](#)). Persistence is the AR(1) coefficient of log income.

## 5.2 Data Moments

I estimate the model by simulated method of moments (SMM), matching seven moments of the consumption and income distribution from the ICRISAT data. The first four moments follow [Bold and Broer \(2021\)](#):

1.  $m_1$ : Variance ratio,  $\text{Var}(\Delta \log c) / \text{Var}(\Delta \log y)$ .
2.  $m_2$ : Regression coefficient,  $\beta(\Delta \log c, \Delta \log y)$ .
3.  $m_3$ : Asymmetric variance ratio,  $[\text{Var}(\Delta \log c \mid \Delta y > 0) - \text{Var}(\Delta \log c \mid \Delta y \leq 0)] / \text{Var}(\Delta \log y)$ .
4.  $m_4$ : Asymmetric beta,  $\beta(\Delta \log c, \Delta \log y \mid \Delta y > 0) - \beta(\Delta \log c, \Delta \log y \mid \Delta y \leq 0)$ .

I augment these with three additional moments that help identify  $\omega$ :

5.  $m_5$ : Skewness of  $\Delta \log c$ .
6.  $m_6$ : Cross-sectional dispersion (standard deviation of  $c_{it}/\bar{c}_t$  across households).
7.  $m_7$ : Persistence of beta differences (correlation of household-specific betas across subperiods).

Table 6 reports the data moments for all three villages with bootstrap standard errors. Several features of the data moments are worth highlighting. The variance ratio  $m_1$  ranges from 0.30 (Aurepalle) to 0.56 (Shirapur), indicating substantial but incomplete consumption smoothing. The regression coefficient  $m_2 \approx 0.17$ – $0.22$  confirms that income has residual predictive power for consumption, consistent with imperfect insurance. The asymmetry moments ( $m_3 < 0$ ,  $m_4 < 0$ ) suggest that insurance is somewhat stronger in bad states than in good states, a pattern that [Bold and Broer \(2021\)](#) show is difficult to generate under standard limited commitment. The cross-sectional dispersion  $m_6$  ranges from 0.15 (Shirapur) to 0.18 (Aurepalle), indicating heterogeneity in consumption levels within villages that the baseline limited commitment model struggles to match.<sup>4</sup>

## 5.3 Estimation Procedure

The model has two free parameters: the discount factor  $\delta$  and the enforcement intensity  $\omega$ , with the CRRA coefficient fixed at  $\rho = 1$  (log utility) throughout. I search over a grid:

- $\delta \in \{0.96, 0.97, 0.98, 0.99\}$
- $\rho = 1$  (log utility)
- $\omega \in \{0, 0.05, 0.10, 0.15, 0.20, 0.30, 0.50, 0.70, 0.85, 0.95\}$

---

<sup>4</sup>The  $m_6$  values in Table 6, which are the estimation targets throughout Sections 5–6, are computed from log per-adult-equivalent consumption. Appendix G (Table G.2) reports  $m_6$  for the VLS1 and VLS2 samples using level-based per-capita consumption following [Morten \(2019\)](#)’s variable definitions, yielding larger values (0.34–0.56 for VLS1). All structural results are estimated against the log-based moments in Table 6.

**Table 6**

Data Moments: ICRISAT Villages

| Moment                                              | Aurepalle         | Shirapur          | Kanzara           |
|-----------------------------------------------------|-------------------|-------------------|-------------------|
| $m_1$ : $\text{Var}(\Delta c)/\text{Var}(\Delta y)$ | 0.304<br>(0.052)  | 0.562<br>(0.180)  | 0.329<br>(0.078)  |
| $m_2$ : $\beta(\Delta c, \Delta y)$                 | 0.206<br>(0.061)  | 0.222<br>(0.070)  | 0.169<br>(0.058)  |
| $m_3$ : Asymmetric var. ratio                       | -0.082<br>(0.068) | -0.248<br>(0.243) | -0.156<br>(0.087) |
| $m_4$ : Asymmetric $\beta$                          | -0.413<br>(0.141) | -0.141<br>(0.164) | -0.053<br>(0.102) |
| $m_5$ : Skewness( $\Delta c$ )                      | -0.408<br>(0.244) | -1.378<br>(1.065) | -0.411<br>(0.301) |
| $m_6$ : XS dispersion                               | 0.179<br>(0.037)  | 0.150<br>(0.023)  | 0.166<br>(0.031)  |
| $m_7$ : Persistence $\beta$ diff.                   | -0.090<br>(0.134) | -0.060<br>(0.160) | 0.032<br>(0.101)  |

*Notes:* Bootstrap standard errors in parentheses, from 1,000 replications resampling households (clusters) with replacement within each village. Moments 1–4 follow (Bold and Broer, 2021); moments 5–7 are additional moments used to identify  $\omega$ . The squares of these standard errors form the diagonal SMM weighting matrix in equation (24).

For each parameter combination and coalition size  $n \in \{2, \dots, n_{\max}\}$ , I solve the model by value function iteration and simulate consumption paths. The SMM criterion is:

$$\mathcal{W}(\theta) = \left[ \hat{m}(\theta) - m^{\text{data}} \right]' \left[ \text{diag}(\hat{V}) \right]^{-1} \left[ \hat{m}(\theta) - m^{\text{data}} \right], \quad (24)$$

where  $\hat{m}(\theta)$  is the vector of simulated moments,  $m^{\text{data}}$  is the vector of data moments, and  $\hat{V}$  is the bootstrap variance-covariance matrix of the data moments (1,000 replications, resampling households with replacement within each village; Table 6). I use the diagonal of  $\hat{V}$  as the weighting matrix, and evaluate the criterion at the equilibrium group size, defined as the largest coalition sustainable under the coalitional deviation structure at each parameter vector.

## 5.4 Results

### 5.4.1 Parameter Estimates

Table 7 reports the SMM estimates for each village under the coalitional deviations (CD) model with position-dependent enforcement. Panel A shows the full 7-moment estimation; Panel B reports a 2-moment estimation using only  $m_1$  and  $m_2$  as a robustness check.

The central estimate is that  $\omega = 0.95$  is selected in the 7-moment estimation uniformly across all three villages: the data consistently prefer position-dependent enforcement. However,  $\omega$  is at the upper boundary of its grid in all villages, and the criterion declines monotonically in  $\omega$  (Figure 2), so the data indicate a *direction*, namely that more position-dependence improves fit,

**Table 7**

SMM Parameter Estimates: CD Model with Position-Dependent Enforcement

|                                     | Aurepalle | Shirapur | Kanzara |
|-------------------------------------|-----------|----------|---------|
| <i>Panel A: 7-moment estimation</i> |           |          |         |
| Group size ( $n^*$ )                | 4         | 3        | 4       |
| Discount factor ( $\delta$ )        | 0.97      | 0.97     | 0.96    |
| Risk aversion ( $\rho$ )            | 1.0       | 1.0      | 1.0     |
| Enforcement ( $\omega$ )            | 0.95      | 0.95     | 0.95    |
| Criterion ( $\mathcal{W}_7$ )       | 15.86     | 17.93    | 12.45   |
| <i>Panel B: 2-moment estimation</i> |           |          |         |
| Discount factor ( $\delta$ )        | 0.97      | 0.98     | 0.96    |
| Enforcement ( $\omega$ )            | 0.85      | 0.95     | 0.95    |
| Criterion ( $\mathcal{W}_2$ )       | 1.95      | 3.31     | 2.59    |

*Notes:* Panel A minimizes the 7-moment criterion  $\mathcal{W}_7$ ; Panel B minimizes  $\mathcal{W}_2$  using only  $m_1$  and  $m_2$ .  $\rho = 1$  (log utility) throughout. Diagonal weighting matrix from the bootstrap variance of the data moments. The criterion is evaluated at the equilibrium (largest sustainable) group size  $n^*$  implied by the coalitional deviation structure at each parameter vector.

rather than a sharply identified level; I return to the identification properties of  $\omega$  in Section 6.4. The discount factor, by contrast, is estimated in the interior of its grid ( $\delta = 0.97$  in Aurepalle and Shirapur, 0.96 in Kanzara), and  $\delta = 0.99$  is never selected. The criterion improvement over the baseline ( $\omega = 0$ ) is substantial, at 46% for Aurepalle, 55% for Shirapur, and 56% for Kanzara, while endogenous group sizes of 3–4 households match Bold and Broer’s findings, so the coalitional deviation structure continues to produce small, stable groups.

#### 5.4.2 Moment Fit

Table 8 compares the data moments with the model at  $\omega = 0$  (Bold and Broer baseline) and the best-fitting  $\omega$  for all three villages.

The main result is a substantial improvement in cross-sectional consumption dispersion ( $m_6$ ). Bold and Broer’s baseline model generates too little variation in consumption across households, leaving everyone too similar. Position-dependent sanctions create heterogeneous outside options, which translate into greater consumption variation in equilibrium. For Aurepalle, the model now matches 82% of observed dispersion (up from 30%); for Kanzara, 61% (up from 17%). Shirapur shows a more modest improvement, matching 51% of observed dispersion (up from 14%). This residual under-prediction is not an artifact of the VLS1 sample: cross-sectional dispersion is at least as large in the VLS2 data (2001–2004; Appendix G).

Table 9 decomposes the criterion into moment-by-moment contributions. The gain is not purchased at the expense of the canonical time-series moments: in Shirapur and Kanzara the best fit selects the same  $\delta$  as the baseline, so the contributions of  $m_1$ – $m_5$  are unchanged by construction, and in Aurepalle the joint contribution of  $m_1$  and  $m_2$  falls (from 3.30 to 1.95) because the

Table 8

Moment Fit: Data vs. Baseline ( $\omega=0$ ) vs. Position-Dependent Enforcement

|       |               | Aurepalle |            |               | Shirapur |            |              | Kanzara |            |              |
|-------|---------------|-----------|------------|---------------|----------|------------|--------------|---------|------------|--------------|
|       | Moment        | Data      | $\omega=0$ | Best          | Data     | $\omega=0$ | Best         | Data    | $\omega=0$ | Best         |
| $m_1$ | Var. ratio    | 0.304     | 0.326      | 0.234         | 0.562    | 0.318      | 0.318        | 0.329   | 0.228      | 0.228        |
| $m_2$ | $\beta$       | 0.206     | 0.314      | <b>0.226</b>  | 0.222    | 0.314      | <b>0.314</b> | 0.169   | 0.225      | <b>0.225</b> |
| $m_3$ | Asym. var     | -0.082    | 0.005      | <b>0.000</b>  | -0.248   | 0.001      | 0.001        | -0.156  | 0.000      | 0.000        |
| $m_4$ | Asym. $\beta$ | -0.413    | 0.003      | <b>-0.004</b> | -0.141   | 0.003      | <b>0.003</b> | -0.053  | -0.001     | -0.001       |
| $m_5$ | Skewness      | -0.408    | 0.014      | <b>-0.008</b> | -1.378   | 0.009      | 0.009        | -0.411  | -0.007     | -0.007       |
| $m_6$ | XS disp.      | 0.179     | 0.054      | <b>0.148</b>  | 0.150    | 0.020      | <b>0.076</b> | 0.166   | 0.028      | <b>0.101</b> |
| $m_7$ | Persist.      | -0.090    | 0.042      | <b>0.009</b>  | -0.060   | 0.026      | <b>0.023</b> | 0.032   | 0.014      | <b>0.031</b> |
|       | Criterion     | —         | 29.5       | <b>15.9</b>   | —        | 39.7       | <b>17.9</b>  | —       | 28.5       | <b>12.4</b>  |
|       | $\delta$      | —         | 0.96       | 0.97          | —        | 0.97       | 0.97         | —       | 0.96       | 0.96         |
|       | $\omega$      | —         | 0          | 0.95          | —        | 0          | 0.95         | —       | 0          | 0.95         |
|       | Improvement   | —         | —          | 46%           | —        | —          | 55%          | —       | —          | 56%          |

Notes: Bold entries indicate improved fit relative to baseline. “Best” minimizes  $\mathcal{W}_7$  over  $(\delta, \omega)$  with the criterion evaluated at the equilibrium group size; it coincides with the  $\omega = 0.95$  row of Table 10. Baseline ( $\omega=0$ ) reproduces the (Bold and Broer, 2021) CD model. In Shirapur and Kanzara the best fit selects the same  $\delta$  as the baseline, so  $m_1$ – $m_5$  are unchanged and the improvement comes entirely from the position-sensitive moments  $m_6$  and  $m_7$ ; in Aurepalle  $\delta$  moves from 0.96 to 0.97.

improvement in the pass-through coefficient  $m_2$  more than offsets a modest deterioration in the variance ratio  $m_1$ . Nor is the improvement confined to  $m_6$ ; the position-dependent pass-through moment  $m_7$  also improves in all three villages.

Asymmetry moments ( $m_3, m_4$ ) move in the correct direction:  $m_4$  becomes negative in Aurepalle and Kanzara, consistent with the data but magnitudes remain small. I discuss this limitation in Section 6.

### 5.4.3 Consumption Dynamics Under Position-Dependent Enforcement

Figure 1 plots simulated consumption changes ( $\Delta \log c$ ) against income changes ( $\Delta \log y$ ) for Aurepalle at the best-fitting parameters. Under perfect insurance, all points would lie on the horizontal axis ( $\Delta \log c = 0$  for all  $\Delta \log y$ ); under autarky, points would cluster along the 45-degree line. The scatter plot shows that the model generates a realistic intermediate pattern: consumption changes are substantially attenuated relative to income changes (partial insurance), but with enough dispersion to match the cross-sectional variation observed in the ICRISAT data. The cloud of points is roughly symmetric around zero, consistent with the near-zero asymmetry moments ( $m_3, m_4$ ).

### 5.4.4 Calibration: How the Criterion Varies with $\omega$

Figure 2 displays the SMM criterion as a function of  $\omega$  for each village, re-optimising  $\delta$  at each grid point. At  $\omega = 0$ , the criterion corresponds exactly to Bold and Broer’s (2021) baseline coalitional deviation model. The criterion falls monotonically with  $\omega$  in all three villages, with the steepest decline at intermediate values of  $\omega$  (around 0.10–0.30) where the additional

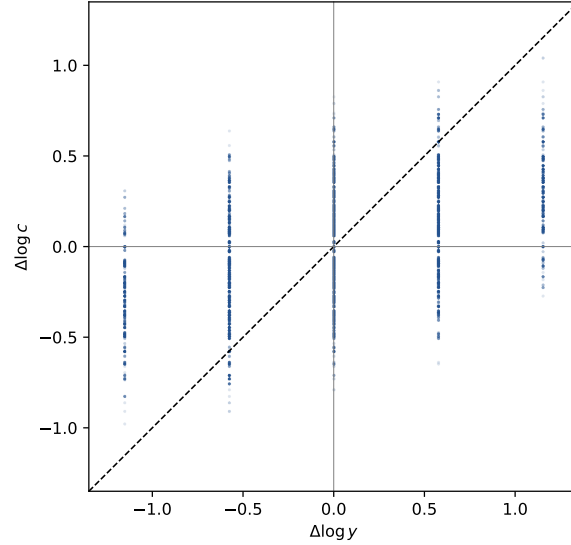


**Table 9**

Criterion Decomposition: Moment-by-Moment Contributions

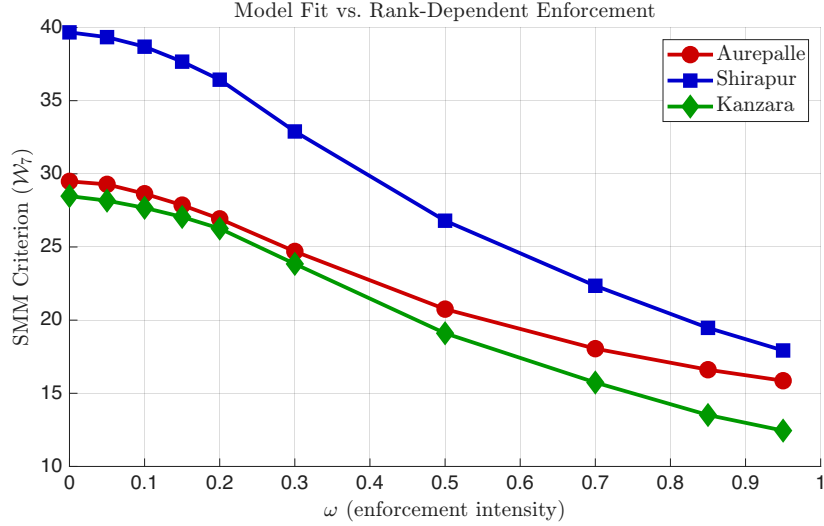
|       |               | Aurepalle  |       | Shirapur   |       | Kanzara    |       |
|-------|---------------|------------|-------|------------|-------|------------|-------|
|       | Moment        | $\omega=0$ | Best  | $\omega=0$ | Best  | $\omega=0$ | Best  |
| $m_1$ | Var. ratio    | 0.18       | 1.84  | 1.85       | 1.85  | 1.67       | 1.67  |
| $m_2$ | $\beta$       | 3.12       | 0.11  | 1.72       | 1.72  | 0.92       | 0.92  |
| $m_3$ | Asym. var     | 1.61       | 1.45  | 1.05       | 1.05  | 3.25       | 3.25  |
| $m_4$ | Asym. $\beta$ | 8.75       | 8.47  | 0.77       | 0.77  | 0.27       | 0.27  |
| $m_5$ | Skewness      | 2.99       | 2.69  | 1.70       | 1.70  | 1.80       | 1.80  |
| $m_6$ | XS disp.      | 11.87      | 0.76  | 32.30      | 10.58 | 20.54      | 4.55  |
| $m_7$ | Persist.      | 0.97       | 0.54  | 0.29       | 0.27  | 0.03       | 0.00  |
| Total |               | 29.48      | 15.86 | 39.67      | 17.93 | 28.48      | 12.45 |

*Notes:* Each cell reports  $(\hat{m}_k - m_k^{\text{data}})^2 / \hat{V}_{kk}$ , the moment's contribution to the SMM criterion (equation 24), at the baseline ( $\omega=0$ ) and best-fit parameters of Table 8. Columns sum to the reported criterion values. The improvement from position-dependent enforcement is concentrated in  $m_6$  (cross-sectional dispersion) and  $m_7$  (position-dependent pass-through); in Shirapur and Kanzara the contributions of  $m_1$ – $m_5$  are unchanged by construction (same  $\delta$ ), and in Aurepalle the joint contribution of the time-series moments  $m_1$  and  $m_2$  falls from 3.30 to 1.95: the model does not purchase cross-sectional fit at the expense of the canonical risk-sharing moments.

**Figure 1**

Simulated Consumption vs. Income Changes for Aurepalle:  $\Delta \log c$  against  $\Delta \log y$  from CD model simulations with position-dependent enforcement at the best-fit parameters ( $\delta = 0.97$ ,  $\omega = 0.95$ ). The dashed line is the 45-degree line (autarky).

cross-sectional dispersion generated by position-dependent sanctions is most valuable for fit. The monotonic decline indicates that the data prefer more position-dependent enforcement than my grid allows; there is no interior optimum within  $\omega \in [0, 0.95]$ . This means  $\omega$  is not sharply identified in the classical sense, the data reveal a direction (more position-dependence improves fit) rather than a precise level. For this reason, I interpret the results as evidence that position-dependent enforcement is quantitatively important, while acknowledging that the exact magnitude of  $\omega$  is uncertain.



**Figure 2**

SMM Criterion ( $\mathcal{W}_7$ ) vs. Enforcement Intensity  $\omega$ . At each  $\omega$ , the discount factor  $\delta$  is re-optimised. The criterion falls monotonically with  $\omega$  in all three villages.

Table 10 reports the full calibration results, showing how each moment evolves as  $\omega$  increases. At each  $\omega$ ,  $\delta$  is re-optimised over the full grid  $\{0.96, 0.97, 0.98, 0.99\}$ ;  $\delta = 0.99$  is never selected. The best fit in Table 8 coincides with the  $\omega = 0.95$  row of Table 10.

The improvement is driven primarily by  $m_6$  (cross-sectional dispersion), which rises steadily with  $\omega$ . Other moments ( $m_1$ – $m_5$ ,  $m_7$ ) are relatively stable across  $\omega$  values, confirming that the enforcement parameter operates primarily through the cross-sectional distribution rather than through time-series properties of individual consumption.

#### 5.4.5 Moment Profiles

Figure 3 displays each of the seven moments as a function of  $\omega$ , with horizontal dashed lines indicating the data targets. The figure reveals the anatomy of the criterion improvement. The variance ratio ( $m_1$ ) and regression coefficient ( $m_2$ ) are relatively stable across  $\omega$  values, confirming that the time-series properties of individual consumption paths are governed primarily by the discount factor  $\delta$  and risk aversion  $\rho$ , not by the enforcement parameter. The asymmetry moments ( $m_3$ ,  $m_4$ ) remain near zero regardless of  $\omega$ , a limitation I discuss in Section 6. The skewness ( $m_5$ ) shows modest sensitivity. The dominant channel is cross-sectional dispersion ( $m_6$ ), which rises steadily with  $\omega$ , closing the gap between the baseline model and the

**Table 10**Calibration: Model Fit by Enforcement Intensity  $\omega$ 

| $\omega$                  | $\delta$ | $\rho$ | $m_1$ | $m_2$ | $m_3$  | $m_4$  | $m_5$  | $m_6$ | $m_7$  | Criterion |
|---------------------------|----------|--------|-------|-------|--------|--------|--------|-------|--------|-----------|
| <i>Panel A: Aurepalle</i> |          |        |       |       |        |        |        |       |        |           |
| Data                      | —        | —      | 0.304 | 0.206 | -0.082 | -0.413 | -0.408 | 0.179 | -0.090 | —         |
| 0.00                      | 0.96     | 1.0    | 0.326 | 0.314 | 0.005  | 0.003  | 0.014  | 0.054 | 0.042  | 29.5      |
| 0.10                      | 0.97     | 1.0    | 0.234 | 0.226 | 0.000  | -0.004 | -0.008 | 0.045 | 0.016  | 28.6      |
| 0.20                      | 0.97     | 1.0    | 0.234 | 0.226 | 0.000  | -0.004 | -0.008 | 0.054 | 0.012  | 26.9      |
| 0.30                      | 0.97     | 1.0    | 0.234 | 0.226 | 0.000  | -0.004 | -0.008 | 0.066 | 0.005  | 24.7      |
| 0.50                      | 0.97     | 1.0    | 0.234 | 0.226 | 0.000  | -0.004 | -0.008 | 0.092 | -0.004 | 20.8      |
| 0.70                      | 0.97     | 1.0    | 0.234 | 0.226 | 0.000  | -0.004 | -0.008 | 0.116 | -0.001 | 18.1      |
| 0.85                      | 0.97     | 1.0    | 0.234 | 0.226 | 0.000  | -0.004 | -0.008 | 0.134 | 0.005  | 16.6      |
| 0.95                      | 0.97     | 1.0    | 0.234 | 0.226 | 0.000  | -0.004 | -0.008 | 0.148 | 0.009  | 15.9      |
| <i>Panel B: Shirapur</i>  |          |        |       |       |        |        |        |       |        |           |
| Data                      | —        | —      | 0.562 | 0.222 | -0.248 | -0.141 | -1.378 | 0.150 | -0.060 | —         |
| 0.00                      | 0.97     | 1.0    | 0.318 | 0.314 | 0.001  | 0.003  | 0.009  | 0.020 | 0.026  | 39.7      |
| 0.10                      | 0.97     | 1.0    | 0.318 | 0.314 | 0.001  | 0.003  | 0.009  | 0.022 | 0.021  | 38.7      |
| 0.20                      | 0.97     | 1.0    | 0.318 | 0.314 | 0.001  | 0.003  | 0.009  | 0.027 | 0.011  | 36.4      |
| 0.30                      | 0.98     | 1.0    | 0.236 | 0.234 | 0.001  | 0.005  | 0.011  | 0.034 | -0.013 | 32.9      |
| 0.50                      | 0.98     | 1.0    | 0.236 | 0.234 | 0.001  | 0.005  | 0.011  | 0.049 | -0.010 | 26.8      |
| 0.70                      | 0.98     | 1.0    | 0.236 | 0.234 | 0.001  | 0.005  | 0.011  | 0.061 | 0.002  | 22.3      |
| 0.85                      | 0.97     | 1.0    | 0.318 | 0.314 | 0.001  | 0.003  | 0.009  | 0.071 | 0.023  | 19.5      |
| 0.95                      | 0.97     | 1.0    | 0.318 | 0.314 | 0.001  | 0.003  | 0.009  | 0.076 | 0.023  | 17.9      |
| <i>Panel C: Kanzara</i>   |          |        |       |       |        |        |        |       |        |           |
| Data                      | —        | —      | 0.329 | 0.169 | -0.156 | -0.053 | -0.411 | 0.166 | 0.032  | —         |
| 0.00                      | 0.96     | 1.0    | 0.228 | 0.225 | 0.000  | -0.001 | -0.007 | 0.028 | 0.014  | 28.5      |
| 0.10                      | 0.96     | 1.0    | 0.228 | 0.225 | 0.000  | -0.001 | -0.007 | 0.030 | 0.013  | 27.7      |
| 0.20                      | 0.96     | 1.0    | 0.228 | 0.225 | 0.000  | -0.001 | -0.007 | 0.035 | 0.010  | 26.3      |
| 0.30                      | 0.97     | 1.0    | 0.176 | 0.174 | -0.001 | -0.002 | -0.002 | 0.050 | -0.014 | 23.8      |
| 0.50                      | 0.97     | 1.0    | 0.176 | 0.174 | -0.001 | -0.002 | -0.002 | 0.071 | -0.018 | 19.1      |
| 0.70                      | 0.97     | 1.0    | 0.176 | 0.174 | -0.001 | -0.002 | -0.002 | 0.089 | -0.011 | 15.7      |
| 0.85                      | 0.97     | 1.0    | 0.176 | 0.174 | -0.001 | -0.002 | -0.002 | 0.103 | -0.007 | 13.5      |
| 0.95                      | 0.96     | 1.0    | 0.228 | 0.225 | 0.000  | -0.001 | -0.007 | 0.101 | 0.031  | 12.4      |

*Notes:* At each  $\omega$ ,  $\delta$  is re-optimized to minimize  $\mathcal{W}_7$  over the grid  $\delta \in \{0.96, 0.97, 0.98, 0.99\}$ ;  $\delta = 0.99$  is never selected. CD model with coalitional deviations; criterion evaluated at the equilibrium group size with diagonal bootstrap-variance weights. The  $\omega = 0.95$  row is the overall best fit reported in Tables 7 and 8.

data. This confirms that the enforcement parameter operates primarily through heterogeneous outside options that translate into cross-sectional consumption variation, rather than through time-series dynamics.

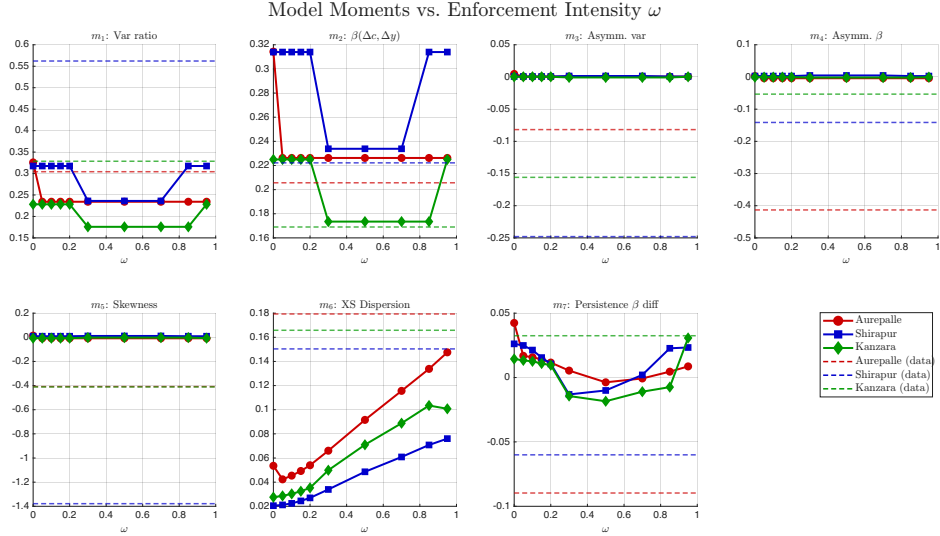


Figure 3

Moment Profiles vs. Enforcement Intensity  $\omega$ . Each panel shows one of the seven moments as a function of  $\omega$ , with dashed horizontal lines marking the data targets. The improvement in fit is driven primarily by  $m_6$  (cross-sectional dispersion).

#### 5.4.6 Robustness: Convex Sanctions

To assess sensitivity to the functional form of sanctions, I estimate a convex alternative:

$$\sigma^{\text{convex}}(y_{it}; \omega) = \omega (1 - r(y_{it}))^2. \quad (25)$$

This concentrates sanctions more heavily on the lowest-income household while going easier on intermediate ranks. Table 11 compares the two specifications, and Figure 4 provides a visual comparison across the full  $\omega$  grid.

Table 11

Robustness: Linear vs. Convex Sanctions ( $\omega=0.95$ )

|                        | Aurepalle    |         | Shirapur     |        | Kanzara      |         |
|------------------------|--------------|---------|--------------|--------|--------------|---------|
|                        | Linear       | Convex  | Linear       | Convex | Linear       | Convex  |
| Criterion              | <b>15.9</b>  | 20.8    | <b>17.9</b>  | 26.8   | <b>12.4</b>  | 19.1    |
| $m_1$ (Var. ratio)     | 0.234        | 0.234   | 0.318        | 0.236  | 0.228        | 0.176   |
| $m_4$ (Asym. $\beta$ ) | -0.0039      | -0.0039 | 0.0028       | 0.0047 | -0.0006      | -0.0018 |
| $m_6$ (XS disp.)       | <b>0.148</b> | 0.092   | <b>0.076</b> | 0.049  | <b>0.101</b> | 0.071   |

Notes: Linear sanctions:  $\sigma = \omega(1 - r)$ . Convex sanctions:  $\sigma = \omega(1 - r)^2$ . Bold indicates better fit.

Linear vs. Convex Rank-Dependent Sanctions

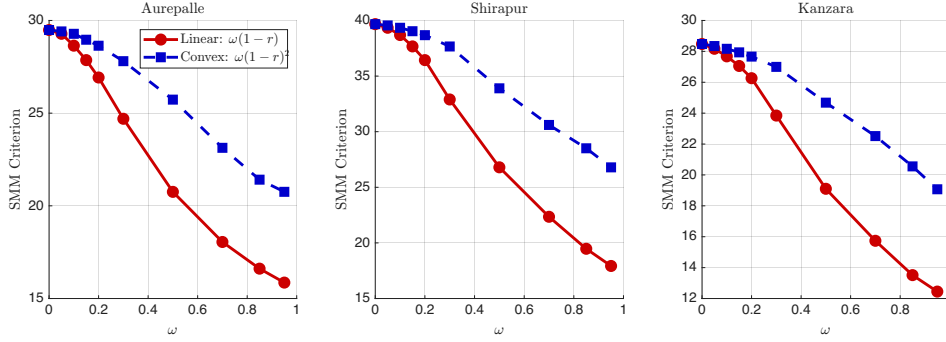


Figure 4

Linear vs. Convex Sanctions: SMM Criterion by Village. Linear sanctions ( $\sigma = \omega(1-r)$ ) dominate convex sanctions ( $\sigma = \omega(1-r)^2$ ) across all villages and all  $\omega$  values.

Linear sanctions strictly dominate across all villages and all  $\omega$  values. The convex specification at  $\omega = 0.95$  performs roughly as well as the linear specification at  $\omega = 0.50$ , confirming that the improvement comes from the degree of differentiation across income states, not from concentrating punishment on the very poorest. This result has a natural interpretation: the data prefer a sanctions function that maintains strong differentiation between intermediate income ranks, not one that loads all punishment onto the bottom of the distribution. The finding is also reassuring for the model’s credibility: it suggests that the improvement in fit comes from a plausible mechanism (graduated social enforcement) rather than from an extreme assumption about the poorest households.

## 6 Discussion

### 6.1 What Drives the Improvement?

The criterion improvement is driven primarily by cross-sectional consumption dispersion ( $m_6$ ). In Bold and Broer’s baseline, all households in a coalition share risk efficiently conditional on the binding pattern of participation constraints. Since the constraints bind relatively symmetrically across income states, equilibrium consumption is too homogeneous relative to the data.

Position-dependent sanctions break this symmetry by creating heterogeneous outside options. Low-income households face severely reduced autarky consumption, which relaxes their participation constraints and allows the planner to offer them less favourable terms. High-income households retain attractive outside options, maintaining binding constraints in good states. The resulting consumption distribution exhibits more cross-sectional variation, closing the gap with the data. The fact that  $\omega$  operates mainly through  $m_6$  while leaving  $m_1$ – $m_5$  relatively unaffected (Table 10, Figure 3) is both a strength and a limitation: it means the enforcement parameter does not distort other moments, but it also means  $\omega$  is identified primarily by a single dimension of the data. I discuss this identification issue further in Section 6.4.

Table 12 makes the mechanism concrete for Aurepalle at the best-fit parameters. Panel A

shows the autarky side: sanctions scale with one minus the rank of the income state, so a household deviating in the low state loses 63% of its autarky consumption, in the middle state 32%, and in the high state nothing. Panel B and Panel C show the equilibrium consequence in the stationary simulation: relaxing the participation constraints of low-rank households allows the group to hold them to persistently less favourable allocations, roughly doubling the cross-sectional standard deviation of log consumption (0.20 to 0.38) and the P90–P10 gap (0.55 to 1.02), and raising the first-order autocorrelation of household log consumption from 0.30 to 0.79. The rise in persistence is where the mechanism leaves its clearest mark, since position operates through slow-moving effective Pareto weights rather than through the current income draw; the same logic explains why conditional means by current income state move comparatively little.

**Table 12**

Worked Example: The Mechanism in Aurepalle ( $\delta = 0.97$ )

|                                                                                                          | Low state | Middle state | High state |
|----------------------------------------------------------------------------------------------------------|-----------|--------------|------------|
| <i>Panel A: Autarky side of the participation constraint</i>                                             |           |              |            |
| Income $y_s$                                                                                             | 0.518     | 0.922        | 1.639      |
| Rank $r(y_s)$                                                                                            | 1/3       | 2/3          | 1          |
| Sanction share $\omega(1 - r(y_s))$ , $\omega = 0.95$                                                    | 63.3%     | 31.7%        | 0.0%       |
| Autarky consumption, $\omega = 0$                                                                        | 0.518     | 0.922        | 1.639      |
| Autarky consumption, $\omega = 0.95$                                                                     | 0.190     | 0.630        | 1.639      |
| <i>Panel B: Equilibrium consumption in the stationary simulation</i>                                     |           |              |            |
| $E[\log c \mid \text{state}]$ , $\omega = 0$                                                             | −0.149    | −0.036       | 0.136      |
| $E[\log c \mid \text{state}]$ , $\omega = 0.95$                                                          | −0.201    | −0.089       | 0.080      |
| <i>Panel C: Cross-sectional distribution of log consumption</i>                                          |           |              |            |
| SD of $\log c$ : 0.200 ( $\omega = 0$ ) vs. 0.384 ( $\omega = 0.95$ )                                    |           |              |            |
| P90–P10 gap: 0.545 ( $\omega = 0$ ) vs. 1.015 ( $\omega = 0.95$ )                                        |           |              |            |
| Persistence $\text{corr}(\log c_t, \log c_{t-1})$ : 0.295 ( $\omega = 0$ ) vs. 0.792 ( $\omega = 0.95$ ) |           |              |            |

*Notes:* Panel A traces the mechanism through the deviation payoff: sanctions scale with one minus the rank of the income state, so a low-state deviator loses 63% of autarky consumption while a high-state deviator loses nothing. Panels B and C show the equilibrium consequence in the stationary simulation at the equilibrium group size ( $n = 4$ ): with  $\omega = 0.95$ , participation constraints of low-rank households are relaxed, allocations tilt against them persistently, and cross-sectional dispersion and persistence of consumption both rise. Conditional means by *current* income state move little because position operates through slow-moving effective Pareto weights rather than the current income draw; the dispersion (Panel C) is the footprint of the mechanism.

## 6.2 Why Asymmetry Remains Small

Despite correct signs on the asymmetry moments ( $m_3$ ,  $m_4$  are negative or near-zero, consistent with the data), magnitudes remain small:  $m_4 \approx -0.001$  versus data targets of  $-0.413$  (Aurepalle) and  $-0.053$  (Kanzara). This limitation is not specific to my extension; Bold and Broer’s baseline generates  $m_4 \approx +0.003$  against the same targets.

The root cause lies in the structure of limited commitment models with CRRA utility. The intraperiod allocation rule, equations (20)–(21), divides resources based on Pareto weights and CRRA marginal utility. This rule is inherently symmetric in its treatment of positive and negative income shocks conditional on the binding pattern. While position-dependent sanctions change *which* constraints bind and how often, they do not change the allocation rule conditional

on a given binding pattern. Generating large asymmetry would require modifying the allocation rule itself, for example, through loss-averse preferences or asymmetric monitoring costs rather than modifying outside options.

### 6.3 Interpretation of $\omega$

The estimated  $\omega = 0.95$  implies that the lowest-income household ( $r = 1/3$  with 3 income states) faces sanctions of  $0.95 \times 2/3 \approx 0.63$ , losing 63% of its income upon deviation. This is large but not implausible in a context where community membership provides access to common resources, credit networks, and social insurance that collectively represent a substantial share of effective income.

The fact that  $\omega$  is near its upper bound across all villages suggests that position-dependent enforcement is quantitatively important for risk-sharing outcomes. However, as discussed below, the boundary solution also raises identification concerns.

### 6.4 Identification of $\omega$

$\omega = 0.95$  is at the upper boundary of the baseline grid in all three villages (Table 7); the discount factor, by contrast, is estimated in the interior ( $\delta = 0.96$ – $0.97$ , with  $\delta = 0.99$  never selected), so the boundary problem is confined to  $\omega$ . The criterion declines monotonically in  $\omega$  (Figure 2), with no interior optimum. Extending the grid confirms this is not an artifact of stopping at 0.95: re-solving the model at  $\omega \in \{0.97, 0.99\}$ , the criterion continues to decline, though at a flattening rate (Aurepalle: 15.86 at  $\omega = 0.95$  to 15.60 at  $\omega = 0.99$ ; Shirapur: 17.93 to 17.49; Kanzara: 12.45 to 12.01). The parameter cannot exceed 1: at  $\omega = 1$  the lowest-ranked deviator’s autarky consumption is zero, which log utility rules out. The estimate is therefore best read as saying that the data prefer the maximal feasible degree of position-dependence, with the criterion nearly flat as  $\omega$  approaches its structural bound. I cannot report a confidence interval for  $\omega$  in the usual sense, and inference at a parameter-space boundary is non-standard (Andrews, 2001); the Monte Carlo exercise below provides a direct finite-sample assessment of what the boundary pattern does and does not imply.

A separate concern is that the criterion improvement is driven predominantly by a single moment,  $m_6$  (cross-sectional dispersion). Table 10 and Figure 3 show that  $m_6$  rises steadily with  $\omega$  while other moments ( $m_1$ – $m_5$ ,  $m_7$ ) are relatively insensitive to  $\omega$ . In this sense,  $\omega$  functions primarily as a cross-sectional dispersion parameter: it is identified mainly by the gap between observed and baseline-predicted consumption heterogeneity.

Table 13 makes the identification argument explicit by reporting the Jacobian of the moment vector with respect to  $(\delta, \omega)$ , computed by finite differences on the estimation grid and scaled by each moment’s standard error. The two parameters move essentially disjoint sets of moments:  $\omega$  has numerically zero effect on  $m_1$ – $m_5$  and moves  $m_6$  by 0.23–0.38 standard errors per grid step, while  $\delta$  moves the time-series moments  $m_1$  and  $m_2$  by 0.5–1.7 standard errors per step. Although a higher  $\delta$  can also raise  $m_6$ , doing so degrades  $m_1$  and  $m_2$  at first order, so the

criterion cannot trade  $\delta$  against  $\omega$ :  $\delta$  is pinned down by the time-series moments and  $\omega$  by the cross-sectional moments conditional on  $\delta$ . Consistent with this, the estimated  $\delta$  is in the interior of its grid in all three villages while only  $\omega$  sits at its boundary.

**Table 13**

Identification: Moment Sensitivity to  $\delta$  and  $\omega$

| Moment                | Aurepalle       |                 | Shirapur        |                 | Kanzara         |                 |
|-----------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
|                       | $\Delta_\delta$ | $\Delta_\omega$ | $\Delta_\delta$ | $\Delta_\omega$ | $\Delta_\delta$ | $\Delta_\omega$ |
| $m_1$ : Var. ratio    | -1.72           | -0.00           | -0.45           | 0.00            | -0.68           | -0.00           |
| $m_2$ : $\beta$       | -1.39           | -0.00           | -1.14           | -0.00           | -0.88           | -0.00           |
| $m_3$ : Asym. var     | 0.00            | -0.00           | 0.00            | 0.00            | -0.01           | 0.00            |
| $m_4$ : Asym. $\beta$ | 0.04            | -0.00           | 0.01            | -0.00           | -0.01           | -0.00           |
| $m_5$ : Skewness      | 0.10            | 0.00            | 0.00            | 0.00            | 0.02            | 0.00            |
| $m_6$ : XS disp.      | 0.58            | 0.38            | -0.30           | 0.23            | 0.36            | 0.33            |
| $m_7$ : Persist.      | -0.03           | 0.03            | -0.12           | 0.00            | -0.37           | 0.03            |

*Notes:* Each entry is the change in the simulated moment from a one-grid-step increase in the parameter, expressed in units of the moment's bootstrap standard error, evaluated at the equilibrium group size around the best-fit point.  $\Delta_\delta$  steps  $\delta$  by 0.01 at  $\omega = 0.95$  (Aurepalle and Shirapur:  $0.97 \rightarrow 0.98$ ; Kanzara:  $0.96 \rightarrow 0.97$ );  $\Delta_\omega$  steps  $\omega$  from 0.85 to 0.95 at the best-fit  $\delta$ . The table shows why the parameters are separately identified.  $\omega$  moves only the position-sensitive moments, with numerically zero effect on  $m_1$ – $m_5$  and an effect of 0.23–0.38 standard errors per step on  $m_6$ ; and although  $\delta$  can also move  $m_6$ , it simultaneously moves the time-series moments  $m_1$  and  $m_2$  by 0.5–1.7 standard errors per step, so the data cannot substitute  $\delta$  for  $\omega$  in fitting  $m_6$  without a first-order deterioration in the canonical risk-sharing moments.  $\delta$  is pinned down by  $m_1$ – $m_2$ , and  $\omega$  by  $m_6$ – $m_7$  conditional on  $\delta$ .

A remaining concern is mechanical: if adding a free parameter that inflates  $m_6$  were always rewarded, the monotone criterion decline would carry no evidential content. I address this with a Monte Carlo experiment (Table 14). I simulate 300 ICRISAT-sized pseudo-panels per village from the *baseline* model ( $\omega = 0$ , best-fit  $\delta$ ) and apply the entire estimation procedure (moment construction, within-panel bootstrap weighting, grid search with  $\delta$  re-optimized at each  $\omega$ ) to each pseudo-panel, and repeat the exercise with data generated at the estimated parameters ( $\omega = 0.95$ ). The procedure has close to exact size and high power. Under the null,  $\hat{\omega} \geq 0.85$  is never selected and the criterion is never monotonically declining in  $\omega$  (0 of 300 replications in every village, Monte Carlo  $p < 0.01$ ); sampling noise produces sizeable criterion improvements, but always at low values of  $\hat{\omega}$ . Under the alternative, the procedure recovers  $\hat{\omega} \geq 0.85$  in 78–100% of replications and reproduces the monotone decline in 57–93%. The configuration observed in the data,  $\hat{\omega} = 0.95$  in all three villages with a monotonically declining criterion, is therefore the signature of the position-dependent enforcement DGP, not an artifact of the estimation design.

This raises the question of whether alternative mechanisms could fit  $m_6$  equally well. I conduct horse races against the two leading candidates: additive consumption measurement error ( $\sigma_\varepsilon$ ), a one-parameter extension that can generate additional cross-sectional dispersion without position-dependent enforcement, and heterogeneous risk aversion, the friction that [Mazzocco and Saini \(2012\)](#) document in these villages and the most natural preference-based competitor.

Table 15 reports the results. The measurement error model (setting  $\omega = 0$  and searching over the



**Table 14**

Monte Carlo: Size and Power of the  $\omega$  Estimation Procedure

| DGP:                                      | Aurepalle    |                 | Shirapur     |                 | Kanzara      |                 |
|-------------------------------------------|--------------|-----------------|--------------|-----------------|--------------|-----------------|
|                                           | $\omega = 0$ | $\omega = 0.95$ | $\omega = 0$ | $\omega = 0.95$ | $\omega = 0$ | $\omega = 0.95$ |
| $P(\hat{\omega} = 0)$                     | 0.35         | 0.00            | 0.37         | 0.00            | 0.40         | 0.00            |
| $P(\hat{\omega} \leq 0.20)$               | 0.87         | 0.00            | 0.99         | 0.00            | 0.98         | 0.00            |
| $P(\hat{\omega} \geq 0.85)$               | 0.00         | 0.98            | 0.00         | 1.00            | 0.00         | 0.78            |
| $P(\text{criterion monotone in } \omega)$ | 0.00         | 0.57            | 0.00         | 0.93            | 0.00         | 0.60            |
| Median criterion improvement (%)          | 6            | 79              | 4            | 80              | 4            | 75              |
| 95th pct. improvement (%)                 | 50           | 93              | 52           | 95              | 55           | 94              |

*Notes:* 300 replications per village and DGP. Each replication draws an ICRISAT-sized pseudo-panel (all simulated agents over a random six-year window) from the model solved at the baseline ( $\omega = 0$ , best-fit  $\delta$ ) or at the estimated parameters ( $\omega = 0.95$ ), then applies the full estimation procedure: seven moments, diagonal weighting matrix from a within-panel cluster bootstrap over households (200 replications), criterion evaluated on the  $(\delta, \omega)$  grid at the equilibrium group size,  $\delta$  re-optimized at each  $\omega$ . “Monotone” means the  $\delta$ -profiled criterion declines at every step of the  $\omega$  grid, the pattern observed in the data (Figure 2). Under the null, the procedure never selects  $\hat{\omega} \geq 0.85$  and the criterion is never monotone (0/300, MC  $p < 0.01$  per village); under the alternative it recovers a high  $\omega$  in 88–100% of replications. The raw improvement percentage alone is a weaker discriminator: in small panels sampling noise can produce large improvements at low  $\hat{\omega}$  under the null (95th percentile 53–60%), which is why the location of  $\hat{\omega}$  and the monotone shape carry the identifying information.

measurement error fraction  $\sigma_\varepsilon$  jointly with  $\delta$ ) reduces the SMM criterion by 13% in Aurepalle, 40% in Shirapur, and 32% in Kanzara. These are substantial improvements, but well below the 46–56% reductions achieved by the sanctions model. Moreover, these improvements come at a cost: because noise is added to log consumption,  $\text{var}(\Delta c)$  increases mechanically, inflating  $m_1$  (the variance ratio). In Shirapur, the best-fit measurement error fraction of 72% inflates the simulated variance ratio to  $m_1 = 0.892$ , far above the data value of 0.562; in Kanzara, 75% noise inflates  $m_1$  to 0.425 against a data value of 0.329. These fractions imply that nearly three-quarters of measured consumption variation would be pure noise which is an implausible characterisation of the ICRISAT panel.

By contrast, position-dependent sanctions improve  $m_6$  *without* distorting  $m_1$ – $m_5$ , because the mechanism operates through outside options rather than through added noise. In Kanzara and Shirapur, the best-fit sanctions model produces identical  $m_1$ – $m_5$  values to the baseline (same  $\delta$ ,  $\rho$ , and group size), with the entire criterion improvement coming from  $m_6$  and  $m_7$ . The sanctions model changes the binding pattern of participation constraints: low-income households face worse outside options and therefore accept less favourable consumption allocations persistently generating cross-sectional dispersion through the structural mechanism rather than through statistical artefacts. This selective improvement in  $m_6$  while preserving the fit of  $m_1$ – $m_5$  is the key identification result: among the alternatives considered, only position-dependent enforcement can account for the observed level of consumption heterogeneity without degrading the model’s ability to match the other risk-sharing diagnostics.

Beyond the horse race itself, the direction of improvement from sanctions is robust across all three villages, and the linear functional form dominates the convex alternative, suggesting the data prefer graduated enforcement across income states over concentrated punishment of the poorest. The baseline model ( $\omega = 0$ ,  $\sigma_\varepsilon = 0$ ) also substantially underpredicts  $m_6$  in all villages,

confirming that some mechanism for additional cross-sectional dispersion is needed. I view these as strong structural findings while acknowledging that the precise level of  $\omega$  is not sharply identified: the boundary solution and monotonically declining criterion mean the data prefer more position-dependence than my grid allows.

**Table 15**

Horse Race: Sanctions vs. Measurement Error

|                                                     | Aurepalle |          |           |            | Shirapur |          |           |            | Kanzara |          |           |            |
|-----------------------------------------------------|-----------|----------|-----------|------------|----------|----------|-----------|------------|---------|----------|-----------|------------|
|                                                     | Data      | Baseline | Sanctions | Meas. Err. | Data     | Baseline | Sanctions | Meas. Err. | Data    | Baseline | Sanctions | Meas. Err. |
| <i>Panel A: Moment fit</i>                          |           |          |           |            |          |          |           |            |         |          |           |            |
| $m_1$ : $\text{Var}(\Delta c)/\text{Var}(\Delta y)$ | 0.304     | 0.326    | 0.234     | 0.328      | 0.562    | 0.318    | 0.318     | 0.892      | 0.329   | 0.228    | 0.228     | 0.425      |
| $m_2$ : $\beta(\Delta c, \Delta y)$                 | 0.206     | 0.314    | 0.226     | 0.226      | 0.222    | 0.314    | 0.314     | 0.235      | 0.169   | 0.225    | 0.225     | 0.095      |
| $m_3$ : Asymm. var. ratio                           | -0.082    | 0.005    | 0.000     | -0.000     | -0.248   | 0.001    | 0.001     | 0.011      | -0.156  | 0.000    | 0.000     | -0.005     |
| $m_4$ : Asymm. $\beta$ diff.                        | -0.413    | 0.003    | -0.004    | -0.005     | -0.141   | 0.003    | 0.003     | 0.003      | -0.053  | -0.001   | -0.001    | -0.000     |
| $m_5$ : Skewness                                    | -0.408    | 0.014    | -0.008    | -0.010     | -1.378   | 0.009    | 0.009     | 0.007      | -0.411  | -0.007   | -0.007    | -0.001     |
| $m_6$ : XS dispersion                               | 0.179     | 0.054    | 0.148     | 0.051      | 0.150    | 0.020    | 0.076     | 0.057      | 0.166   | 0.028    | 0.101     | 0.064      |
| $m_7$ : Position $\beta$ diff.                      | -0.090    | 0.042    | 0.009     | 0.017      | -0.060   | 0.026    | 0.023     | 0.008      | 0.032   | 0.014    | 0.031     | -0.002     |
| <i>Panel B: Criterion and parameters</i>            |           |          |           |            |          |          |           |            |         |          |           |            |
| SMM criterion                                       |           | 29.48    | 15.86     | 25.78      |          | 39.67    | 17.93     | 23.74      |         | 28.48    | 12.45     | 19.45      |
| Improvement (%)                                     |           |          | 46.2      | 12.6       |          |          | 54.8      | 40.2       |         |          | 56.3      | 31.7       |
| $\delta$                                            |           | 0.96     | 0.97      | 0.97       |          | 0.97     | 0.97      | 0.98       |         | 0.96     | 0.96      | 0.99       |
| $\hat{\omega}$                                      |           | —        | 0.95      | —          |          | —        | 0.95      | —          |         | —        | 0.95      | —          |
| Meas. err. fraction                                 |           | —        | —         | 0.21       |          | —        | —         | 0.72       |         | —        | —         | 0.75       |
| Group size                                          |           | 3        | 4         | 4          |          | 3        | 3         | 4          |         | 4        | 4         | 8          |

*Notes:* Each column reports the best-fit model from grid search over  $\delta$  and the respective extension parameter ( $\omega$  for sanctions,  $\sigma_\varepsilon$  for measurement error), minimising the 7-moment SMM criterion with diagonal bootstrap-variance weights, evaluated at the equilibrium (largest sustainable) group size. Baseline sets  $\omega = 0$  and  $\sigma_\varepsilon = 0$ . Improvement is the percentage reduction in the criterion relative to baseline. Moments  $m_1$ – $m_4$  are standard risk-sharing diagnostics;  $m_5$  (skewness),  $m_6$  (cross-sectional dispersion), and  $m_7$  (position-dependent pass-through) are the three additional moments targeting  $\omega$  identification.

The second horse race asks whether heterogeneous risk aversion can substitute for position-dependent enforcement. I solve [Bold and Broer](#)’s heterogeneous-preference specification of the limited commitment model, in which half the village has low and half has high CRRA curvature, at spreads ranging from none to the extreme  $\rho_i \in \{0.3, 1.7\}$ , with the discount factor re-optimized at each spread.<sup>5</sup> Table 16 reports the results: risk-aversion heterogeneity does not generate the missing dispersion. Across all three villages and all spreads,  $m_6$  never exceeds 0.079, against data values of 0.150–0.179, and even the extreme spread leaves  $m_6$  close to its homogeneous-preference level (Aurepalle 0.069 to 0.068, Shirapur 0.021 to 0.020, Kanzara 0.037 to 0.036). The reason is that with modest aggregate risk, heterogeneous curvature reallocates exposure to aggregate shocks across types but creates little *persistent* dispersion in consumption levels, which is what  $m_6$  measures and what heterogeneous outside options deliver. Since  $\omega$  is identified through  $m_6$ , this rules out the concern that the enforcement parameter is absorbing risk-aversion heterogeneity. Heterogeneous discount factors, which accumulate Pareto-weight dispersion over time and are therefore a more potent dispersion mechanism, remain untested and are noted as a limitation in Section 7.

<sup>5</sup>The heterogeneous-preference model is available under individual deviations, so its criterion levels are not comparable to the coalitional-deviation tables: the individual-deviations baseline is strongly rejected on the asymmetry and skewness moments regardless of preferences, which is [Bold and Broer](#)’s original finding. The informative comparison is how the moments respond to the preference spread within the fixed framework. The published replication code for this specification omits the within-period allocation step of the simulation; I implement it from the model’s first-order conditions (consumption solves  $c_i = (\kappa a_i)^{-1/\rho_i}$  with  $\kappa$  set by the group resource constraint), which reduces to the published homogeneous updating rule when the types coincide.

**Table 16**

Heterogeneous Risk Aversion as an Alternative to Position-Dependent Enforcement

| Heterogeneity             | Aurepalle |       |       |                 | Shirapur |       |       |                 | Kanzara |       |       |                 |
|---------------------------|-----------|-------|-------|-----------------|----------|-------|-------|-----------------|---------|-------|-------|-----------------|
|                           | $m_1$     | $m_2$ | $m_6$ | $\mathcal{W}_7$ | $m_1$    | $m_2$ | $m_6$ | $\mathcal{W}_7$ | $m_1$   | $m_2$ | $m_6$ | $\mathcal{W}_7$ |
| Data                      | 0.304     | 0.206 | 0.179 | —               | 0.562    | 0.222 | 0.150 | —               | 0.329   | 0.169 | 0.166 | —               |
| none ( $\rho_i = 1$ )     | 0.188     | 0.294 | 0.069 | 178.6           | 0.140    | 0.255 | 0.021 | 64.6            | 0.131   | 0.247 | 0.037 | 110.4           |
| $\rho_i \in \{0.7, 1.3\}$ | 0.231     | 0.328 | 0.079 | 173.7           | 0.095    | 0.188 | 0.018 | 61.9            | 0.006   | 0.020 | 0.006 | 79.2            |
| $\rho_i \in \{0.5, 1.5\}$ | 0.211     | 0.291 | 0.071 | 168.0           | 0.088    | 0.159 | 0.019 | 59.8            | 0.120   | 0.199 | 0.034 | 102.9           |
| $\rho_i \in \{0.3, 1.7\}$ | 0.219     | 0.270 | 0.068 | 164.9           | 0.115    | 0.167 | 0.020 | 58.1            | 0.152   | 0.207 | 0.036 | 97.9            |

Notes: (Bold and Broer, 2021) limited commitment model with heterogeneous CRRA risk aversion (two types, half the village each) and no position-dependent enforcement ( $\omega = 0$ ), solved under individual deviations, which is the specification for which the heterogeneous-preference model is available. At each spread, the discount factor is chosen from  $\{0.80, 0.82, \dots, 0.92\}$  to minimize the 7-moment criterion with the empirical bootstrap weights. Moments are computed from the stationary simulated panel.

## 6.5 Relation to the Reduced-Form Evidence

The structural results are consistent with the reduced-form findings in Section 3. The reduced-form regressions show that rank is a level predictor of consumption dynamics, while the interaction of income rank with income shocks is not significant. The structural model provides one specific mechanism consistent with this pattern: position-dependent enforcement creates heterogeneous outside options, so that low-ranked households face relaxed participation constraints (their outside option is unattractive), leading to systematically different consumption allocations. The reduced-form finding that the position effect operates through levels rather than through differential shock pass-through matches the structural mechanism, in which position affects the *binding pattern* of participation constraints (a level effect on allocations) rather than the marginal sensitivity of the allocation rule to income realisations. This claim has limits. The reduced-form significance of rank is consistent with any mechanism that induces a correlation between income rank and consumption levels, including preferences over relative position (Section 3); the structural evidence favours the enforcement channel only among the alternatives explicitly tested. A sharper check is to run the augmented Townsend regression of Section 3 on panels simulated from the model at both  $\omega = 0$  and  $\omega = 0.95$  (Table 17), and here the result is a genuine limitation: income rank is almost never significant in the simulated regressions under *either* DGP. The model’s position channel manifests as persistent consumption-level dispersion, which is why  $\omega$  is identified through  $m_6$  and  $m_7$ ; but with a symmetric three-state income process, current income rank is too coarse and too transitory to enter the differenced Townsend regression the way the empirical rank measure does. The structural results should therefore be read as evidence that position-dependent enforcement explains the *cross-sectional moments* the baseline model misses, not as a full structural account of the reduced-form rank coefficient, part of which plausibly reflects persistent household heterogeneity (in income levels, preferences, or both) outside the model.

**Table 17**

Augmented Townsend Regressions on Model-Simulated Panels

|                                       | Aurepalle    |                 | Shirapur     |                 | Kanzara      |                 |
|---------------------------------------|--------------|-----------------|--------------|-----------------|--------------|-----------------|
| DGP:                                  | $\omega = 0$ | $\omega = 0.95$ | $\omega = 0$ | $\omega = 0.95$ | $\omega = 0$ | $\omega = 0.95$ |
| Mean $\hat{\zeta}$ (rank coefficient) | 0.023        | 0.018           | 0.013        | 0.009           | 0.018        | 0.024           |
| P( $\hat{\zeta}$ significant at 5%)   | 0.01         | 0.01            | 0.02         | 0.01            | 0.02         | 0.02            |
| P(interaction significant at 5%)      | 0.11         | 0.10            | 0.06         | 0.08            | 0.07         | 0.07            |

*Notes:* 300 pseudo-panels per village and DGP (all simulated agents over random six-year windows). Approach 1 regression of consumption growth on the cross-sectional mean of consumption growth, own income growth, and income rank  $RP_{it}^y$ ; Approach 2 adds  $RP_{it}^y \times \Delta y_{it}$ . Standard errors clustered by household. Rank is essentially never significant under either DGP: the model’s position channel operates through persistent consumption-level dispersion ( $m_6, m_7$ ), which the differenced Townsend regression with a coarse three-state income rank does not pick up. The empirical rank significance in Table 3 therefore reflects variation, namely persistent income differences across households, that the model’s symmetric three-state income process abstracts from.

## 6.6 Welfare Implications

The estimated model can be used to quantify what position-dependent enforcement costs the village in welfare terms. I simulate the stationary economy at the best-fit discount factor and equilibrium group size, varying only  $\omega$ , and compare per-period ex-ante welfare  $E[\log c]$  across economies; because aggregate consumption is pinned down by the resource constraint and is identical across  $\omega$ , any welfare difference is purely distributional. Table 18 reports the results.

The welfare cost of the estimated enforcement regime is economically meaningful: moving from  $\omega = 0$  to the estimated  $\omega = 0.95$  lowers ex-ante welfare by the equivalent of 5.2% of consumption in Aurepalle, 2.6% in Shirapur, and 3.4% in Kanzara. This is the price, in ex-ante terms, of the additional consumption inequality that position-dependent sanctions sustain in equilibrium, with the cross-sectional variance of log consumption roughly quadrupling relative to the position-independent benchmark. The cost is also close to linear in  $\omega$ . Halving the enforcement parameter removes 50–55% of the welfare loss, so an intervention that improves the outside options of low-ranked households, such as a guaranteed-employment scheme or index insurance that does not depend on social standing, need not eliminate position-dependence entirely to recover much of the loss. Finally, the incidence conditional on the *current* income state is nearly uniform, a direct implication of the model’s mechanism: sanctions distort allocations through the history of binding participation constraints, embodied in slow-moving effective Pareto weights, so the households who bear the cost are those with persistently weak positions rather than those with a currently low income draw.

The exercise is a steady-state comparison rather than a transition analysis, holding the income process and group structure fixed at their estimated values and varying only the enforcement technology. It also abstracts from any disciplining role that position-dependent sanctions might play in sustaining participation in the first place, since it removes sanctions in a world where limited commitment is the only friction; within the model this margin is captured by the equilibrium group size, which barely changes across  $\omega$  at the estimated  $\delta$ .

**Table 18**

Welfare Cost of Position-Dependent Enforcement

|                                                                                      | Aurepalle | Shirapur | Kanzara |
|--------------------------------------------------------------------------------------|-----------|----------|---------|
| <i>Consumption-equivalent welfare change relative to <math>\omega = 0</math> (%)</i> |           |          |         |
| $\omega = 0.50$                                                                      | -2.62     | -1.31    | -1.53   |
| $\omega = 0.95$ (estimated)                                                          | -5.21     | -2.63    | -3.43   |
| <i>Cross-sectional variance of log consumption</i>                                   |           |          |         |
| $\omega = 0$                                                                         | 0.040     | 0.020    | 0.028   |
| $\omega = 0.50$                                                                      | 0.092     | 0.047    | 0.059   |
| $\omega = 0.95$                                                                      | 0.148     | 0.076    | 0.101   |
| Share of welfare cost removed by halving $\omega$ (%)                                | 50        | 50       | 55      |

*Notes:* Stationary simulated economies at the best-fit discount factor ( $\delta = 0.97, 0.97, 0.96$ ) and equilibrium group size, varying only  $\omega$ . Mean aggregate consumption is identical across  $\omega$  by the resource constraint, so the welfare differences are purely distributional. With log utility and ex-ante identical households, per-period ex-ante welfare is  $E[\log c]$ ; the consumption-equivalent change is  $\exp\{E[\log c | \omega] - E[\log c | \omega = 0]\} - 1$ . Conditional on the current income state, losses are close to uniform (within 0.4 percentage points across states), because a household's position in the model is history-dependent through the evolution of its effective Pareto weight rather than tied to its current income draw.

## 6.7 Connection to Rank-Dependent Expected Utility

There is a deep formal connection between my framework and the rank-dependent expected utility (RDEU) literature initiated by (Quiggin, 1982) and (Yaari, 1987). In the RDEU framework, preferences are represented by:

$$\text{RDEU}(X) = \int u(x) \varphi'(1 - F(x)) f(x) dx, \quad (26)$$

where  $\varphi : [0, 1] \rightarrow [0, 1]$  is a probability distortion function. The term  $\varphi'(1 - F(x))$  reweights the density according to the *rank* of outcome  $x$  in the distribution  $F$ . When  $\varphi$  is convex (“pessimism”), low outcomes receive disproportionate weight, generating risk aversion.

In (Yaari, 1987)’s dual theory, the special case with  $u(x) = x$ , risk aversion comes entirely from distorting probabilities rather than from curvature of the utility function. For a binary gamble with outcomes  $\{y^L, y^H\}$  and  $\Pr(y^H) = p$ , the certainty equivalent is  $y^L + \varphi(p)(y^H - y^L)$ . The risk premium is:

$$\pi = \frac{p - \varphi(p)}{1 - p} (y^H - y^L), \quad (27)$$

which is positive under pessimism ( $\varphi(p) < p$ ) and, unlike the Arrow-Pratt approximation, is *first-order* in the stake size rather than second-order.

The connection to my reduced-form results is direct. In equation (3), the term  $1 - F(c_{it})$  that enters the modified first-order condition is precisely the survival function, the same object that appears in the RDEU Choquet integral (26). My Approach 2 (equation 5), in which the planner’s FOC features  $[1 - F(U_i(c_{it}))]$  in the denominator, mirrors the structure of RDEU with a nonlinear distortion applied to the distribution of utilities. Put differently, my reduced-form finding, that  $1 - F(c)$  belongs in the Townsend test, is observationally consistent with

households evaluating risk-sharing arrangements through rank-dependent probability weights. Despite this parallel, I model the enforcement channel rather than the preference channel. RDEU preferences are generally time-inconsistent in dynamic settings: a plan optimal ex ante need not be optimal when re-evaluated at a future node, because the probability distortion is applied to the ex-ante rather than the conditional distribution, and in a limited commitment framework where households re-evaluate participation every period, time inconsistency creates fundamental difficulties for the definition of self-enforcing arrangements. Moreover, [Chateauneuf et al. \(2000\)](#) show that when all agents share the same convex probability distortion, Pareto-optimal risk sharing under RDEU is identical to that under expected utility, so departures require heterogeneous distortion functions, adding parameters and sacrificing tractability. Position-dependent enforcement, by contrast, has direct micro-foundations in the anthropological literature on village sanctions ([Fafchamps, 2011](#); [Kaur, 2019](#)), while the assumption that Indian villagers could have probability-distorted preferences is a stronger empirical assertion best left for future work.

My approach can therefore be understood as capturing the empirical content of RDEU: position-dependent treatment of agents and improved cross-sectional fit, through a more parsimonious and computationally tractable channel available, modifying outside options within a standard expected utility framework, rather than replacing the preference apparatus entirely.

## 7 Conclusion

I document that relative position is a robust correlate of risk-sharing outcomes in Indian village economies. A household’s rank in the within-group distribution is a predictor of consumption dynamics across all villages and castes in the ICRISAT data. The finding holds with both consumption rank (directly motivated by a preference-based theoretical derivation) and income rank, which is free of mechanical correlation and bridges directly to the structural model’s income-state rank. Controlling for rank substantially reduces or eliminates rejections of efficient risk sharing.

To quantify this finding structurally, I extend [Bold and Broer \(2021\)](#)’s limited commitment model with income-state-dependent enforcement of insurance arrangements. The data consistently prefer this extension: the SMM criterion improves by 46–56% across three villages, driven primarily by a substantial improvement in the model’s ability to match cross-sectional consumption dispersion. Core findings on small stable groups and approximately symmetric insurance are preserved.

Several limitations remain. The model does not resolve the asymmetric insurance puzzle: consumption responses to positive and negative shocks remain approximately symmetric, while the data suggest slightly asymmetric responses, and this appears to be a structural limitation of CRRA utility within limited commitment rather than of my extension. The enforcement intensity parameter  $\omega$  is at the boundary of my grid in all villages, and the criterion declines monotonically in  $\omega$ , so the precise level is not sharply identified; the data reveal a direction, more

position-dependence improves fit, rather than a precise magnitude. And while the horse races in Section 6.4 show that sanctions dominate the two leading alternatives (measurement error inflates  $m_1$  counterfactually, and heterogeneous risk aversion, tested up to the extreme spread  $\rho_i \in \{0.3, 1.7\}$ , leaves cross-sectional dispersion nearly unchanged, Table 16), one preference-based alternative remains untested: heterogeneous discount factors, which accumulate Pareto-weight dispersion over time and could in principle generate persistent consumption inequality through patience rather than enforcement. Distinguishing patience heterogeneity from position-dependent enforcement, ideally in a specification that nests both, is the natural next step.

Several extensions are possible in future work. Most ambitiously, the formal connection identified in Section 6 between my framework and rank-dependent expected utility Yaari (1987) and Quiggin (1982) suggests a preference-based alternative. Replacing CRRA with RDEU preferences, or, in the limiting case, Yaari’s dual theory with linear utility and nonlinear probability weights would move risk aversion from the curvature of  $u(\cdot)$  to distortion of the cumulative distribution. Under cumulative prospect theory (Tversky and Kahneman, 1992), separate probability distortion for gains and losses could break the symmetry of the intraperiod allocation rule, potentially resolving the asymmetric insurance puzzle ( $m_3$ ,  $m_4$ ) that my model inherits from CRRA. Optimal risk sharing under dual utility takes a layered or tranche structure rather than the smooth proportional sharing implied by the Borch rule Boonen (2017), which may better describe informal insurance arrangements. However, embedding RDEU preferences in a dynamic limited commitment model with coalitional deviations raises deep technical challenges such as time inconsistency, Choquet integration within the Bellman equation, and the neutrality result of Chateauneuf et al. (2000) under homogeneous distortion that remain open. Within the expected utility paradigm, loss-averse preferences Köszegi and Rabin (2006) could generate large asymmetries by making the allocation rule state-dependent. Asymmetric monitoring where bad shocks are easier to verify than good ones could provide an alternative enforcement-based foundation for asymmetric insurance. And combining position-dependent enforcement with preference heterogeneity Chiappori et al. (2014) could jointly address the risk-sharing puzzle at both village and caste levels.

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## Appendix A Robustness of the Reduced-Form Test

The main text reports reduced-form results using both consumption rank (Table 2) and income rank (Table 3). This appendix reports the full set of robustness tests using consumption rank  $1 - F(c_{it})$  across different consumption categories and testing methods. Tables A.1–A.3 report these results. Following (Shrinivas and Fafchamps, 2018), I address the concern that inference can be sensitive to the choice of specification by implementing both the (Townsend, 1994) specification (deviation from village/caste average as the dependent variable) and the (Deaton, 1990) specification (time dummies). I consider three consumption categories: total expenditure (grain, food, nondurables, and leisure), total expenditure excluding leisure, and grain and food only. For each combination, I estimate the standard test, Approach 1 (equation 7), and Approach 2 (equation 8).

The results are consistent across specifications. Relative position is significant at the 1% level in virtually every specification and consumption category. The key finding, that Approach 2 eliminates or substantially reduces rejections of efficient risk sharing, is robust to the choice of testing method and consumption measure.<sup>6</sup> The summary statistics for all specifications are reported in Table 4 in the main text. Following (Mazzocco and Saini, 2012), I also note that among the eight castes identified as having heterogeneous preferences, spurious rejection due to preference heterogeneity could have affected at most 1 of 9 castes under any method, including my approaches.

## Appendix B GMM Estimation

As a robustness check, I re-estimate Approach 2 using the generalized method of moments (GMM). I estimate the Approach 2 moment conditions:

$$E[(\Delta c_{it} - \alpha - \zeta_1 \Delta y_{it} - \zeta_2 RP_{vc} - \gamma RP_{vc} \times \Delta y_{it}) \otimes Z_{it}] = 0,$$

where the instrument set is  $Z_{it} = \{1, \Delta y_{it}, RP_{vc}, RP_{vc} \times \Delta y_{it}, (\Delta y_{it})^2, RP_{vc}^2, \Delta y_{i,t-1}\}$ : seven instruments for four parameters, providing three overidentifying restrictions. I use iterated two-step efficient GMM with the (Hansen, 1982) optimal weighting matrix.

Table B.1 reports the GMM and OLS estimates side by side. The key findings are qualitatively robust: relative position remains highly significant under GMM in all three villages, with coefficients of the same order of magnitude as OLS. The income coefficient remains small and insignificant in Shirapur and Kanzara under both estimators, and becomes insignificant under GMM in Aurepalle.

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<sup>6</sup>For brevity, Tables A.1–A.3 report the coefficients on nonlabour income and relative position but omit the interaction term  $\gamma$  from Approach 2 (equation 8). The interaction term  $RP_{vc} \times \Delta y_{it}$  is generally small and statistically insignificant across specifications, consistent with the main effect of relative position operating additively rather than through a differential income pass-through.

Table A.1

Standard Test using Total Expenditure and Different Testing Methods

| Unit      | Townsend            | Approach 1          |                        | Approach 2          |                        | Deaton              | Approach 1          |                         | Approach 2          |                         | N    |
|-----------|---------------------|---------------------|------------------------|---------------------|------------------------|---------------------|---------------------|-------------------------|---------------------|-------------------------|------|
|           | Nonlabour<br>Income | Nonlabour<br>Income | Relative<br>Position   | Nonlabour<br>Income | Relative<br>Position   | Nonlabour<br>Income | Nonlabour<br>Income | Relative<br>Position    | Nonlabour<br>Income | Relative<br>Position    |      |
| Aurepalle | 0.015***<br>(0.004) | 0.011***<br>(0.003) | -67.644***<br>(1.932)  | 0.007<br>(0.008)    | -67.622***<br>(1.932)  | 0.019***<br>(0.004) | 0.013***<br>(0.004) | -87.232***<br>(2.039)   | 0.010<br>(0.008)    | -87.207***<br>(2.040)   | 3635 |
| Madiga    | 0.045<br>(0.145)    | 0.042<br>(0.126)    | -49.796***<br>(3.624)  | 0.113<br>(0.210)    | -49.840***<br>(3.628)  | 0.019<br>(0.178)    | -0.038<br>(0.109)   | -100.186***<br>(3.582)  | 0.090<br>(0.180)    | -100.288***<br>(3.585)  | 589  |
| Mala      | -0.019<br>(0.126)   | 0.002<br>(0.115)    | -53.350***<br>(7.713)  | -0.229<br>(0.197)   | -52.712***<br>(7.708)  | -0.044<br>(0.272)   | -0.122<br>(0.177)   | -139.684***<br>(11.160) | -0.295<br>(0.255)   | -138.579***<br>(11.226) | 238  |
| Kurba     | 0.021<br>(0.021)    | -0.007<br>(0.017)   | -48.773***<br>(2.828)  | -0.004<br>(0.025)   | -48.814***<br>(2.843)  | 0.036<br>(0.029)    | 0.014<br>(0.018)    | -83.839***<br>(2.926)   | 0.004<br>(0.024)    | -83.712***<br>(2.934)   | 595  |
| Gowda     | 0.033***<br>(0.011) | 0.018*<br>(0.009)   | -82.807***<br>(3.975)  | 0.035**<br>(0.017)  | -83.350***<br>(3.998)  | 0.036**<br>(0.014)  | 0.019**<br>(0.009)  | -128.374***<br>(4.156)  | 0.036**<br>(0.016)  | -128.854***<br>(4.172)  | 944  |
| Reddi     | 0.005<br>(0.003)    | 0.004<br>(0.003)    | -28.868***<br>(2.856)  | 0.004<br>(0.005)    | -28.871***<br>(2.859)  | 0.011**<br>(0.005)  | 0.007*<br>(0.004)   | -53.793***<br>(3.629)   | 0.008<br>(0.006)    | -53.799***<br>(3.635)   | 595  |
| Brahmin   | -0.002<br>(0.004)   | -0.002<br>(0.004)   | -12.428***<br>(3.267)  | -0.005<br>(0.007)   | -12.400***<br>(3.272)  | -0.004<br>(0.011)   | -0.001<br>(0.010)   | -32.662***<br>(7.064)   | -0.002<br>(0.013)   | -32.493***<br>(7.147)   | 238  |
| Kanzara   | 0.009**<br>(0.004)  | 0.006*<br>(0.003)   | -126.395***<br>(2.741) | 0.006<br>(0.007)    | -126.395***<br>(2.742) | 0.009*<br>(0.005)   | 0.002<br>(0.003)    | -151.979***<br>(2.817)  | 0.000<br>(0.006)    | -151.985***<br>(2.818)  | 3677 |
| Mahar     | 0.003<br>(0.012)    | 0.002<br>(0.010)    | -94.858***<br>(6.836)  | 0.033**<br>(0.015)  | -93.996***<br>(6.800)  | 0.004<br>(0.014)    | 0.000<br>(0.011)    | -150.033***<br>(8.018)  | 0.031**<br>(0.015)  | -148.636***<br>(7.981)  | 611  |
| Mali      | 0.027**<br>(0.013)  | 0.014<br>(0.011)    | -107.550***<br>(4.064) | 0.018<br>(0.020)    | -107.576***<br>(4.068) | 0.030**<br>(0.015)  | -0.001<br>(0.009)   | -164.619***<br>(3.929)  | 0.010<br>(0.017)    | -164.680***<br>(3.930)  | 1376 |
| Gosavi    | 0.003<br>(0.006)    | 0.002<br>(0.005)    | -54.367***<br>(7.823)  | 0.017<br>(0.013)    | -54.890***<br>(7.827)  | 0.006<br>(0.009)    | 0.004<br>(0.007)    | -107.514***<br>(11.300) | 0.008<br>(0.017)    | -107.373***<br>(11.341) | 321  |
| Maratha   | 0.005<br>(0.006)    | 0.000<br>(0.005)    | -118.442***<br>(5.859) | -0.005<br>(0.008)   | -118.595***<br>(5.864) | 0.004<br>(0.007)    | -0.000<br>(0.005)   | -148.975***<br>(6.388)  | -0.008<br>(0.008)   | -149.300***<br>(6.391)  | 738  |
| Shirapur  | 0.003<br>(0.003)    | -0.000<br>(0.002)   | -113.868***<br>(2.833) | 0.003<br>(0.004)    | -113.889***<br>(2.833) | 0.004<br>(0.003)    | 0.002<br>(0.002)    | -136.355***<br>(2.835)  | 0.008**<br>(0.004)  | -136.459***<br>(2.834)  | 3243 |
| Mahar     | 0.261***<br>(0.094) | 0.049<br>(0.081)    | -67.683***<br>(5.214)  | 0.133<br>(0.113)    | -68.054***<br>(5.225)  | 0.440***<br>(0.137) | 0.128<br>(0.092)    | -111.816***<br>(5.603)  | 0.116<br>(0.119)    | -111.772***<br>(5.619)  | 423  |
| Dhangar   | 0.006<br>(0.004)    | 0.002<br>(0.003)    | -97.670***<br>(4.953)  | 0.001<br>(0.005)    | -97.672***<br>(4.956)  | 0.008*<br>(0.005)   | 0.006*<br>(0.003)   | -123.535***<br>(4.992)  | 0.007<br>(0.005)    | -123.531***<br>(4.995)  | 818  |
| Maratha   | 0.000<br>(0.003)    | -0.004<br>(0.003)   | -111.725***<br>(4.480) | -0.007<br>(0.008)   | -111.723***<br>(4.481) | 0.000<br>(0.004)    | -0.002<br>(0.003)   | -146.935***<br>(4.538)  | -0.002<br>(0.007)   | -146.941***<br>(4.541)  | 1576 |

Notes: Standard errors in parentheses. \*p<0.1; \*\*p<0.05; \*\*\*p<0.01. Bonferroni corrections applied. Townsend specification uses deviation from village/caste average as dependent variable; Deaton specification uses time dummies.

Table A.2

Standard Test using Total Expenditure (no leisure) and Different Testing Methods

| Unit      | Townsend             | Approach 1           |                       | Approach 2           |                       | Deaton               | Approach 1           |                       | Approach 2           |                       | N    |
|-----------|----------------------|----------------------|-----------------------|----------------------|-----------------------|----------------------|----------------------|-----------------------|----------------------|-----------------------|------|
|           | Nonlabour<br>Income  | Nonlabour<br>Income  | Relative<br>Position  | Nonlabour<br>Income  | Relative<br>Position  | Nonlabour<br>Income  | Nonlabour<br>Income  | Relative<br>Position  | Nonlabour<br>Income  | Relative<br>Position  |      |
| Aurepalle | 0.008***<br>(0.002)  | 0.007***<br>(0.002)  | -5.434***<br>(1.241)  | 0.004<br>(0.005)     | -5.417***<br>(1.241)  | 0.009***<br>(0.003)  | 0.009***<br>(0.002)  | -6.354***<br>(1.387)  | 0.006<br>(0.005)     | -6.328***<br>(1.388)  | 3646 |
| Madiga    | -0.029***<br>(0.009) | -0.033***<br>(0.011) | -0.519<br>(0.338)     | -0.050***<br>(0.017) | -0.512<br>(0.338)     | -0.038***<br>(0.011) | -0.044***<br>(0.014) | -0.713<br>(0.490)     | -0.068***<br>(0.022) | -0.704<br>(0.489)     | 592  |
| Mala      | 0.002<br>(0.008)     | 0.003<br>(0.007)     | -1.678***<br>(0.497)  | -0.000<br>(0.013)    | -1.669***<br>(0.498)  | 0.005<br>(0.017)     | 0.003<br>(0.017)     | -3.870***<br>(1.043)  | 0.001<br>(0.024)     | -3.854***<br>(1.053)  | 238  |
| Kurba     | -0.003<br>(0.004)    | -0.004<br>(0.004)    | -1.382**<br>(0.677)   | -0.002<br>(0.006)    | -1.403**<br>(0.680)   | -0.005<br>(0.006)    | -0.005<br>(0.006)    | -2.156**<br>(0.958)   | -0.003<br>(0.008)    | -2.178**<br>(0.961)   | 595  |
| Gowda     | 0.005<br>(0.003)     | 0.004<br>(0.003)     | -3.934***<br>(1.383)  | 0.002<br>(0.006)     | -3.868***<br>(1.392)  | 0.006<br>(0.004)     | 0.005<br>(0.004)     | -5.778***<br>(1.797)  | 0.003<br>(0.007)     | -5.695***<br>(1.805)  | 947  |
| Reddi     | 0.005**<br>(0.002)   | 0.004*<br>(0.002)    | -11.655***<br>(2.143) | 0.006<br>(0.004)     | -11.677***<br>(2.144) | 0.010***<br>(0.004)  | 0.008**<br>(0.003)   | -20.162***<br>(3.097) | 0.011**<br>(0.005)   | -20.214***<br>(3.100) | 595  |
| Brahmin   | 0.002<br>(0.006)     | 0.001<br>(0.006)     | -10.537**<br>(4.830)  | -0.006<br>(0.010)    | -10.474**<br>(4.832)  | 0.004<br>(0.016)     | 0.007<br>(0.016)     | -28.619**<br>(11.112) | -0.004<br>(0.021)    | -27.465**<br>(11.210) | 238  |
| Kanzara   | 0.003**<br>(0.001)   | 0.003**<br>(0.001)   | -8.034***<br>(1.035)  | 0.003<br>(0.003)     | -8.034***<br>(1.035)  | 0.004**<br>(0.001)   | 0.003**<br>(0.001)   | -9.575***<br>(1.146)  | 0.003<br>(0.003)     | -9.576***<br>(1.146)  | 3701 |
| Mahar     | -0.005<br>(0.003)    | -0.005<br>(0.003)    | -2.108<br>(2.079)     | -0.008*<br>(0.004)   | -2.170<br>(2.080)     | -0.006<br>(0.004)    | -0.005<br>(0.004)    | -2.458<br>(2.692)     | -0.009*<br>(0.005)   | -2.583<br>(2.693)     | 619  |
| Mali      | 0.004<br>(0.003)     | 0.005<br>(0.003)     | -4.433***<br>(1.194)  | 0.000<br>(0.006)     | -4.418***<br>(1.194)  | 0.005<br>(0.004)     | 0.005<br>(0.004)     | -6.507***<br>(1.480)  | 0.000<br>(0.007)     | -6.490***<br>(1.480)  | 1383 |
| Gosavi    | 0.000<br>(0.003)     | -0.000<br>(0.003)    | -20.737***<br>(3.942) | 0.004<br>(0.007)     | -20.892***<br>(3.951) | 0.000<br>(0.004)     | -0.000<br>(0.004)    | -37.392***<br>(6.349) | 0.000<br>(0.010)     | -37.360***<br>(6.373) | 321  |
| Maratha   | 0.005<br>(0.003)     | 0.003<br>(0.003)     | -8.702***<br>(3.296)  | 0.006<br>(0.005)     | -8.629***<br>(3.298)  | 0.006*<br>(0.004)    | 0.005<br>(0.003)     | -11.328***<br>(3.979) | 0.008<br>(0.005)     | -11.227***<br>(3.983) | 742  |
| Shirapur  | 0.001<br>(0.001)     | 0.001<br>(0.001)     | -7.269***<br>(1.094)  | 0.002<br>(0.002)     | -7.282***<br>(1.093)  | 0.001<br>(0.001)     | 0.001<br>(0.001)     | -8.730***<br>(1.174)  | 0.002<br>(0.002)     | -8.756***<br>(1.174)  | 3269 |
| Mahar     | 0.017<br>(0.012)     | 0.015<br>(0.013)     | -0.371<br>(0.790)     | 0.030*<br>(0.018)    | -0.417<br>(0.790)     | 0.026<br>(0.018)     | 0.027<br>(0.019)     | -0.367<br>(1.101)     | 0.045*<br>(0.025)    | -0.425<br>(1.102)     | 425  |
| Dhangar   | 0.003**<br>(0.001)   | 0.003**<br>(0.001)   | -5.876***<br>(1.669)  | 0.004**<br>(0.002)   | -5.880***<br>(1.669)  | 0.004***<br>(0.001)  | 0.003**<br>(0.001)   | -7.556***<br>(1.931)  | 0.005**<br>(0.002)   | -7.549***<br>(1.931)  | 830  |
| Maratha   | -0.000<br>(0.001)    | -0.001<br>(0.001)    | -10.194***<br>(1.661) | -0.005*<br>(0.003)   | -10.181***<br>(1.660) | -0.000<br>(0.001)    | -0.001<br>(0.001)    | -13.055***<br>(1.876) | -0.005<br>(0.003)    | -12.992***<br>(1.875) | 1587 |

Notes: Standard errors in parentheses. \*p&lt;0.1; \*\*p&lt;0.05; \*\*\*p&lt;0.01. Bonferroni corrections applied.

Table A.3

Standard Test using Expenditure on Grain and Food only and Different Testing Methods

| Unit      | Approach 1           |                      |                       | Approach 2           |                       | Deaton               | Approach 1           |                       |                      | Approach 2            |  | N    |
|-----------|----------------------|----------------------|-----------------------|----------------------|-----------------------|----------------------|----------------------|-----------------------|----------------------|-----------------------|--|------|
|           | Nonlabour<br>Income  | Nonlabour<br>Income  | Relative<br>Position  | Nonlabour<br>Income  | Relative<br>Position  | Nonlabour<br>Income  | Nonlabour<br>Income  | Relative<br>Position  | Nonlabour<br>Income  | Relative<br>Position  |  |      |
| Aurepalle | 0.006**<br>(0.002)   | 0.005**<br>(0.002)   | -5.183***<br>(1.173)  | 0.002<br>(0.005)     | -5.168***<br>(1.173)  | 0.006**<br>(0.003)   | 0.006**<br>(0.002)   | -6.076***<br>(1.310)  | 0.003<br>(0.005)     | -6.053***<br>(1.311)  |  | 3646 |
| Madiga    | -0.027***<br>(0.008) | -0.032***<br>(0.010) | -0.520<br>(0.321)     | -0.055***<br>(0.017) | -0.511<br>(0.321)     | -0.036***<br>(0.011) | -0.041***<br>(0.013) | -0.790*<br>(0.465)    | -0.072***<br>(0.021) | -0.778*<br>(0.463)    |  | 592  |
| Mala      | -0.001<br>(0.007)    | -0.000<br>(0.007)    | -1.396***<br>(0.479)  | -0.005<br>(0.012)    | -1.382***<br>(0.481)  | -0.002<br>(0.017)    | -0.004<br>(0.016)    | -3.260***<br>(1.014)  | -0.009<br>(0.023)    | -3.229***<br>(1.023)  |  | 238  |
| Kurba     | -0.007*<br>(0.004)   | -0.008**<br>(0.004)  | -1.540**<br>(0.634)   | -0.009*<br>(0.006)   | -1.520**<br>(0.637)   | -0.011**<br>(0.005)  | -0.012**<br>(0.005)  | -2.407***<br>(0.897)  | -0.013*<br>(0.007)   | -2.390***<br>(0.899)  |  | 595  |
| Gowda     | 0.002<br>(0.003)     | 0.001<br>(0.003)     | -3.652***<br>(1.344)  | -0.001<br>(0.006)    | -3.595***<br>(1.353)  | 0.003<br>(0.004)     | 0.002<br>(0.004)     | -5.329***<br>(1.747)  | -0.000<br>(0.007)    | -5.258***<br>(1.756)  |  | 947  |
| Reddi     | 0.004*<br>(0.002)    | 0.003<br>(0.002)     | -11.294***<br>(1.984) | 0.006<br>(0.003)     | -11.322***<br>(1.985) | 0.008**<br>(0.003)   | 0.006**<br>(0.003)   | -19.519***<br>(2.867) | 0.010**<br>(0.005)   | -19.588***<br>(2.869) |  | 595  |
| Brahmin   | -0.003<br>(0.004)    | -0.004<br>(0.004)    | -8.512***<br>(3.074)  | -0.007<br>(0.007)    | -8.488***<br>(3.079)  | -0.012<br>(0.010)    | -0.009<br>(0.010)    | -22.216***<br>(7.046) | -0.010<br>(0.013)    | -22.184***<br>(7.131) |  | 238  |
| Kanzara   | 0.002**<br>(0.001)   | 0.002*<br>(0.001)    | -8.010***<br>(0.909)  | -0.001<br>(0.002)    | -8.010***<br>(0.909)  | 0.003**<br>(0.001)   | 0.002*<br>(0.001)    | -9.488***<br>(1.006)  | -0.001<br>(0.002)    | -9.497***<br>(1.005)  |  | 3701 |
| Mahar     | -0.005*<br>(0.003)   | -0.005*<br>(0.003)   | -3.264*<br>(1.757)    | -0.007**<br>(0.004)  | -3.322*<br>(1.757)    | -0.006*<br>(0.003)   | -0.006*<br>(0.003)   | -3.769<br>(2.294)     | -0.009**<br>(0.004)  | -3.880*<br>(2.296)    |  | 619  |
| Mali      | 0.001<br>(0.003)     | 0.001<br>(0.003)     | -4.571***<br>(0.986)  | -0.007<br>(0.005)    | -4.546***<br>(0.986)  | 0.001<br>(0.003)     | 0.001<br>(0.003)     | -6.934***<br>(1.220)  | -0.008<br>(0.005)    | -6.905***<br>(1.219)  |  | 1383 |
| Gosavi    | 0.001<br>(0.003)     | 0.001<br>(0.003)     | -20.479***<br>(3.926) | 0.005<br>(0.007)     | -20.641***<br>(3.934) | 0.001<br>(0.004)     | 0.001<br>(0.004)     | -36.529***<br>(6.329) | 0.002<br>(0.010)     | -36.488***<br>(6.353) |  | 321  |
| Maratha   | 0.002<br>(0.002)     | 0.001<br>(0.002)     | -8.080***<br>(2.827)  | -0.001<br>(0.004)    | -8.125***<br>(2.830)  | 0.003<br>(0.003)     | 0.002<br>(0.003)     | -10.499***<br>(3.408) | 0.000<br>(0.005)     | -10.554***<br>(3.412) |  | 742  |
| Shirapur  | 0.000<br>(0.001)     | 0.000<br>(0.001)     | -6.703***<br>(0.998)  | 0.001<br>(0.001)     | -6.711***<br>(0.998)  | 0.000<br>(0.001)     | 0.000<br>(0.001)     | -7.945***<br>(1.071)  | 0.002<br>(0.001)     | -7.964***<br>(1.072)  |  | 3269 |
| Mahar     | 0.023*<br>(0.012)    | 0.023*<br>(0.012)    | 0.056<br>(0.756)      | 0.047***<br>(0.017)  | -0.022<br>(0.754)     | 0.036**<br>(0.017)   | 0.036**<br>(0.018)   | 0.208<br>(1.052)      | 0.067***<br>(0.023)  | 0.111<br>(1.048)      |  | 425  |
| Dhangar   | 0.002**<br>(0.001)   | 0.002**<br>(0.001)   | -5.602***<br>(1.513)  | 0.003**<br>(0.001)   | -5.606***<br>(1.513)  | 0.003**<br>(0.001)   | 0.003**<br>(0.001)   | -7.175***<br>(1.749)  | 0.004**<br>(0.002)   | -7.168***<br>(1.749)  |  | 830  |
| Maratha   | -0.000<br>(0.001)    | -0.001<br>(0.001)    | -9.807***<br>(1.485)  | -0.005**<br>(0.003)  | -9.794***<br>(1.484)  | -0.001<br>(0.001)    | -0.001<br>(0.001)    | -12.323***<br>(1.678) | -0.006**<br>(0.003)  | -12.255***<br>(1.677) |  | 1587 |

Notes: Standard errors in parentheses. \*p&lt;0.1; \*\*p&lt;0.05; \*\*\*p&lt;0.01. Bonferroni corrections applied.

Table B.1

GMM Estimation of the Risk-Sharing Test with Relative Position

|                        | Aurepalle             |                       | Shirapur             |                      | Kanzara              |                      |
|------------------------|-----------------------|-----------------------|----------------------|----------------------|----------------------|----------------------|
|                        | OLS                   | GMM                   | OLS                  | GMM                  | OLS                  | GMM                  |
| $\Delta y$ (income)    | 0.0292***<br>(0.0090) | 0.0336***<br>(0.0092) | -0.0016<br>(0.0065)  | -0.0041<br>(0.0052)  | 0.0045<br>(0.0085)   | 0.0058<br>(0.0084)   |
| Relative position (RP) | 38.084***<br>(2.245)  | 29.231***<br>(1.912)  | 56.446***<br>(3.383) | 45.452***<br>(2.956) | 56.160***<br>(3.173) | 44.231***<br>(2.878) |
| $RP \times \Delta y$   | -0.0179<br>(0.0172)   | -0.0227<br>(0.0176)   | 0.0194<br>(0.0134)   | 0.0238**<br>(0.0104) | 0.0126<br>(0.0178)   | 0.0153<br>(0.0172)   |
| Hansen $J$ -statistic  | —                     | 71.302                | —                    | 47.873               | —                    | 79.975               |
| $p$ -value             | —                     | [0.000]               | —                    | [0.000]              | —                    | [0.000]              |
| $N$                    | 3604                  | 3604                  | 3212                 | 3212                 | 3642                 | 3642                 |

Notes: OLS uses Approach 2 specification with robust (HC1) standard errors. GMM uses iterated two-step efficient GMM with instruments  $\{1, \Delta y, RP, RP \times \Delta y, (\Delta y)^2, RP^2, \Delta y_{t-1}\}$ . Hansen  $J$ -test: null hypothesis is that the overidentifying restrictions are valid. Year-month fixed effects included. \*p<0.1; \*\*p<0.05; \*\*\*p<0.01.

**Important caveats.** This GMM exercise has limitations that the reader should bear in mind. First, the instrument set includes contemporaneous  $RP_{vc}$  and  $RP_{vc} \times \Delta y_{it}$  as their own instruments. If  $RP$  is endogenous due to its mechanical relationship with  $\Delta c$  (as discussed in Section 3), including it in the instrument set does not address that endogeneity. The only genuinely predetermined instrument in  $Z_{it}$  is  $\Delta y_{i,t-1}$ ; the remaining instruments provide efficiency gains but not identification. Second, the Hansen  $J$ -statistic rejects the overidentifying restrictions in all three villages. While  $J$ -test over-rejection in large panels is well documented, rejection here is also consistent with instrument invalidity or model misspecification and should not be dismissed lightly. For these reasons, I view this exercise as confirming that the qualitative patterns are not driven by OLS-specific small-sample properties, but I do not interpret the GMM estimates as providing causal identification. The lagged-rank regressions in Appendix E and the IV exercise in Appendix D, which use genuinely predetermined instruments, provide stronger evidence on this front.

## Appendix C Relative Position and Transfer Flows

My structural model implies that relative position affects risk-sharing outcomes through enforcement: low-ranked households face harsher sanctions for deviating from the insurance arrangement. If this mechanism operates through the transfer system, I should observe that relative position predicts the direction and magnitude of transfers within villages and castes.

I construct transfer variables from the ICRISAT Household Transaction Schedule following (Mazzocco and Saini, 2012). Net transfers are defined as transfers received minus transfers sent (per capita). Within-caste transfers include gifts and loans exchanged with members of the same caste group.

Table C.1 reports the results. Higher-ranked households (those with higher consumption) both receive and send significantly more transfers in all three villages (Panel C), consistent with higher-ranked households being more active participants in the informal insurance system, though also consistent with scale effects or network centrality and so not uniquely attributable to the enforcement channel. In Shirapur, net transfers are significantly *negatively* associated with rank (Panel A): higher-ranked households are net senders, while lower-ranked households are net receivers. This pattern is broadly consistent with the insurance motive. However, the net transfer result is significant only in Shirapur and not in the other two villages, limiting its generality. The effect persists after controlling for nonlabour income (Panel B), suggesting that rank captures something beyond current income, but I interpret these transfer patterns as broadly consistent with the model rather than as direct evidence for the enforcement mechanism specifically.



**Table C.1**

## Relative Position and Transfer Flows

|                                                                          | Aurepalle            | Shirapur             | Kanzara             |
|--------------------------------------------------------------------------|----------------------|----------------------|---------------------|
| <i>Panel A: Dependent variable = Net transfers received (per capita)</i> |                      |                      |                     |
| Relative position (RP)                                                   | -3.950<br>(9.132)    | -7.024**<br>(2.732)  | 4.653<br>(3.667)    |
| <i>Panel B: Net transfers, controlling for nonlabour income</i>          |                      |                      |                     |
| Relative position (RP)                                                   | -4.076<br>(9.188)    | -6.667**<br>(2.662)  | 4.687<br>(3.687)    |
| <i>Panel C: Direction of transfers</i>                                   |                      |                      |                     |
| RP → Transfers received                                                  | 19.168***<br>(6.625) | 5.619***<br>(2.148)  | 8.768**<br>(3.499)  |
| RP → Transfers sent                                                      | 23.118***<br>(7.103) | 12.643***<br>(2.382) | 4.115***<br>(1.357) |
| <i>Panel D: Within-caste net transfers</i>                               |                      |                      |                     |
| RP (caste) → Net caste transfers                                         | 2.150<br>(4.242)     | -3.760<br>(2.970)    | 8.341**<br>(3.580)  |
| <i>N</i>                                                                 | 4350                 | 3792                 | 4050                |

*Notes:* Transfer variables constructed from ICRISAT Household Transaction Schedule following (Mazzocco and Saini, 2012). Net transfers = transfers received – transfers sent (per capita). Within-caste transfers include gifts and loans to/from same-caste members. All regressions include year-month fixed effects. Robust (HC1) standard errors in parentheses. \*p<0.1; \*\*p<0.05; \*\*\*p<0.01.

## Appendix D Measurement Error Corrections

A concern in the risk-sharing literature is that measurement error in consumption or income can bias inference (Ravallion and Chaudhuri, 1997; Mazzocco and Saini, 2012). Classical measurement error in the income variable (on the right-hand side) attenuates its coefficient toward zero, potentially generating false non-rejection of full insurance. Classical measurement error in consumption (on the left-hand side) inflates the residual variance without biasing coefficients, but nonclassical measurement error, correlated with income or relative position, could bias inference in either direction. I address these concerns through three complementary approaches.

**Table D.1**

Measurement Error Corrections: IV and Overidentification Tests

|                                                                | Aurepalle             |                       | Shirapur             |                      | Kanzara              |                      |
|----------------------------------------------------------------|-----------------------|-----------------------|----------------------|----------------------|----------------------|----------------------|
|                                                                | OLS                   | 2SLS                  | OLS                  | 2SLS                 | OLS                  | 2SLS                 |
| <i>Panel A: IV for income using lagged income (Approach A)</i> |                       |                       |                      |                      |                      |                      |
| $\Delta y$ (income)                                            | 0.0222**<br>(0.0089)  | −0.0021<br>(0.0355)   | −0.0059<br>(0.0057)  | −0.0589*<br>(0.0326) | 0.0035<br>(0.0091)   | −0.0027<br>(0.0428)  |
| Relative position (RP)                                         | 33.755***<br>(2.301)  | 33.887***<br>(2.310)  | 53.847***<br>(3.434) | 54.191***<br>(3.513) | 49.577***<br>(2.995) | 49.360***<br>(3.059) |
| First-stage $F$                                                | —                     | 44.2                  | —                    | 1.5                  | —                    | 9.8                  |
| <i>Panel B: Forward vs. reverse regression (Approach B)</i>    |                       |                       |                      |                      |                      |                      |
|                                                                | Forward               | Reverse               | Forward              | Reverse              | Forward              | Reverse              |
| Income/Consumption coeff.                                      | 0.0169***<br>(0.0043) | 0.2486***<br>(0.0667) | 0.0038<br>(0.0031)   | 0.1642<br>(0.1228)   | 0.0083<br>(0.0051)   | 0.1047<br>(0.0660)   |
| Relative position (RP)                                         | 34.050***<br>(2.295)  | 2.499<br>(6.546)      | 53.535***<br>(3.424) | 7.437<br>(20.734)    | 49.433***<br>(2.981) | 8.087<br>(10.220)    |
| <i>Panel C: Overidentification test (Approach C)</i>           |                       |                       |                      |                      |                      |                      |
| F-test (lagged income, rain, landholding)                      | 4.141 [0.001]         |                       | 1.512 [0.182]        |                      | 6.634 [0.000]        |                      |
| Lagged $\Delta y$ coefficient                                  | 0.0009 (0.0036)       |                       | 0.0039 (0.0034)      |                      | −0.0026 (0.0041)     |                      |
| Rainfall coefficient                                           | 0.1918*** (0.0451)    |                       | 0.1253*** (0.0394)   |                      | −0.0058 (0.0309)     |                      |
| $N$                                                            | 3604                  |                       | 3212                 |                      | 3642                 |                      |

*Notes:* Panel A instruments  $\Delta y$  with lagged  $\Delta y$  following (Townsend, 1994) and (Mazzocco and Saini, 2012). Panel B reports the forward regression ( $\Delta c$  on  $\Delta y$ ) and the reverse regression ( $\Delta y$  on  $\Delta c$ ) following (Ravallion and Chaudhuri, 1997). Panel C tests the overidentifying restrictions of full insurance conditional on relative position: under the null, lagged income, rainfall, and landholding class should not predict consumption changes after controlling for  $\Delta y$ , RP, and year-month fixed effects.  $F$ -statistic  $p$ -values in brackets. Robust (HC1) standard errors in parentheses. \* $p < 0.1$ ; \*\* $p < 0.05$ ; \*\*\* $p < 0.01$ .

Table D.1 reports the results. Panel A instruments current income changes with lagged income changes following (Townsend, 1994) and (Mazzocco and Saini, 2012). If measurement error in income attenuates its coefficient, 2SLS should produce a larger (in absolute value) income coefficient. In all three villages, the 2SLS income coefficient is smaller and insignificant, the opposite of what classical measurement error would predict. Meanwhile, the relative position coefficient is virtually unchanged between OLS and 2SLS, confirming that the significance of RP is not an artifact of correlated measurement error. The first-stage  $F$ -statistic is strong for Aurepalle (44.2) and adequate for Kanzara (9.8), though weak for Shirapur (1.5), where the income process is closest to i.i.d.

Panel B implements the reverse regression following (Ravallion and Chaudhuri, 1997): if mea-

surement error in consumption drives the results, then regressing income on consumption (rather than consumption on income) should reveal different patterns, since the bias operates in opposite directions. The forward regression shows income significant in Aurepalle and insignificant elsewhere; the reverse regression shows consumption predicting income in Aurepalle and marginally elsewhere. Crucially, relative position is highly significant in the forward regression but loses significance in the reverse regression, consistent with RP being a genuine determinant of consumption dynamics rather than an artifact of measurement error.

Panel C tests the overidentifying restrictions of full insurance conditional on relative position. Under the null that insurance is efficient after controlling for relative position and income, pre-determined variables such as lagged income, landholding class, and rainfall should not predict consumption changes. The  $F$ -test rejects in Aurepalle and Kanzara but not in Shirapur. Examining individual variables, rainfall is the main source of rejection in Aurepalle, while lagged income and landholding class are individually insignificant in all villages. The relative position coefficient is unaffected by the inclusion of these additional controls, confirming its robustness.

## Appendix E Lagged Income Rank

To further support the reduced-form results, I replace the contemporaneous income rank  $RP_{it}^y = 1 - F_y(y_{it})$  with its one-period lag  $RP_{i,t-1}^y = 1 - F_y(y_{i,t-1})$ . Although income rank is already free of the mechanical correlation that plagues consumption rank (since  $RP^y$  is constructed from income, not consumption), the lagged specification provides an additional check: it confirms that the income-rank effect is not driven by contemporaneous joint determination of income and consumption.

**Table E.1**  
Robustness: Lagged Income Rank

|                             | Aurepalle             |                       |                       |                      | Shirapur              |                      |                       |                       | Kanzara                |                      |                        |                      |
|-----------------------------|-----------------------|-----------------------|-----------------------|----------------------|-----------------------|----------------------|-----------------------|-----------------------|------------------------|----------------------|------------------------|----------------------|
|                             | Approach 1            |                       | Approach 2            |                      | Approach 1            |                      | Approach 2            |                       | Approach 1             |                      | Approach 2             |                      |
|                             | Contemp.<br>(1)       | Lagged<br>(2)         | Contemp.<br>(3)       | Lagged<br>(4)        | Contemp.<br>(5)       | Lagged<br>(6)        | Contemp.<br>(7)       | Lagged<br>(8)         | Contemp.<br>(9)        | Lagged<br>(10)       | Contemp.<br>(11)       | Lagged<br>(12)       |
| $\Delta y^{nl}$             | 0.0137***<br>(0.0031) | 0.0136***<br>(0.0031) | 0.0087<br>(0.0069)    | 0.0157**<br>(0.0064) | 0.0059*<br>(0.0032)   | 0.0087**<br>(0.0038) | 0.0056<br>(0.0047)    | 0.0113***<br>(0.0041) | 0.0103***<br>(0.0029)  | 0.0061*<br>(0.0033)  | 0.0083<br>(0.0076)     | -0.0001<br>(0.0071)  |
| Income rank ( $RP^y$ )      | -53.693***<br>(6.416) | 18.209***<br>(3.525)  | -53.707***<br>(6.414) | 18.170***<br>(3.521) | -85.181***<br>(8.203) | 22.717***<br>(4.839) | -85.179***<br>(8.208) | 22.685***<br>(4.849)  | -92.664***<br>(10.860) | 20.941***<br>(3.784) | -92.684***<br>(10.877) | 21.085***<br>(3.794) |
| $RP^y \times \Delta y^{nl}$ |                       |                       | 0.0094<br>(0.0092)    | -0.0043<br>(0.0108)  |                       |                      | 0.0008<br>(0.0069)    | -0.0071<br>(0.0069)   |                        |                      | 0.0045<br>(0.0159)     | 0.0126<br>(0.0144)   |
| Controls                    | Yes                   | Yes                   | Yes                   | Yes                  | Yes                   | Yes                  | Yes                   | Yes                   | Yes                    | Yes                  | Yes                    | Yes                  |
| $N$                         | 3635                  | 3635                  | 3635                  | 3635                 | 3243                  | 3243                 | 3243                  | 3243                  | 3677                   | 3677                 | 3677                   | 3677                 |
| $R^2$                       | 0.287                 | 0.156                 | 0.287                 | 0.156                | 0.293                 | 0.123                | 0.293                 | 0.123                 | 0.334                  | 0.159                | 0.334                  | 0.159                |

Notes: “Contemp.” uses  $RP_v^y = 1 - F_y(y_{it})$ ; “Lagged” uses  $RP_{v,t-1}^y = 1 - F_y(y_{i,t-1})$ . All regressions include MS controls. Standard errors clustered at household level. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table E.1 reports the results. Lagged income rank produces *positive*, highly significant coefficients (+18 to +21,  $p < 0.001$  in all villages), indicating that households with higher past income experience larger subsequent consumption growth. This is consistent with the structural model’s prediction that high-income households retain favourable insurance terms through position-dependent enforcement. The interaction  $RP_{t-1}^y \times \Delta y^{nl}$  is not significant in any village, consistent with the contemporaneous result that the position effect operates through levels

rather than differential shock sensitivity.

## Consumption Rank: Lagged and Leave-One-Out

For completeness, Table E.2 reports lagged and leave-one-out specifications using *consumption* rank. The leave-one-out measure yields coefficients virtually identical to the original, confirming that the mechanical effect of a single observation on the ECDF is negligible with 40 households per village. Lagged consumption rank produces a positive, highly significant coefficient, the opposite sign from the contemporaneous measure. This sign reversal reflects the mechanical relationship between contemporaneous consumption rank and  $\Delta c$ : it disappears entirely when income rank is used instead (Table 3).

**Table E.2**  
Robustness: Lagged and Leave-One-Out Relative Position

|                                   | Aurepalle             |                       |                       | Shirapur               |                        |                        | Kanzara                |                       |                        |
|-----------------------------------|-----------------------|-----------------------|-----------------------|------------------------|------------------------|------------------------|------------------------|-----------------------|------------------------|
|                                   | Orig.<br>(1)          | Lagged<br>(2)         | LOO<br>(3)            | Orig.<br>(4)           | Lagged<br>(5)          | LOO<br>(6)             | Orig.<br>(7)           | Lagged<br>(8)         | LOO<br>(9)             |
| $\Delta \bar{c}_v$                | 0.1378***<br>(0.0281) | 0.2485***<br>(0.0288) | 0.1378***<br>(0.0281) | 0.1015***<br>(0.0223)  | 0.2065***<br>(0.0302)  | 0.1016***<br>(0.0223)  | 0.1703***<br>(0.0197)  | 0.2883***<br>(0.0241) | 0.1703***<br>(0.0197)  |
| $\Delta y^{nl}$                   | 0.0060<br>(0.0063)    | 0.0167**<br>(0.0084)  | 0.0060<br>(0.0063)    | 0.0108***<br>(0.0041)  | 0.0174***<br>(0.0033)  | 0.0108***<br>(0.0041)  | 0.0098<br>(0.0069)     | 0.0008<br>(0.0084)    | 0.0098<br>(0.0069)     |
| $w_{t-1}$                         | -3.923***<br>(0.634)  | -0.660**<br>(0.282)   | -3.923***<br>(0.634)  | -4.407***<br>(0.449)   | -0.853**<br>(0.361)    | -4.408***<br>(0.449)   | -5.442***<br>(0.471)   | -1.310***<br>(0.269)  | -5.444***<br>(0.471)   |
| $n_{hh}$                          | 3.301***<br>(0.723)   | -1.789***<br>(0.281)  | 3.302***<br>(0.723)   | 7.309***<br>(1.493)    | -0.990<br>(0.631)      | 7.310***<br>(1.493)    | 9.306***<br>(1.387)    | -1.103<br>(0.714)     | 9.309***<br>(1.388)    |
| Interview freq.                   | 0.048<br>(0.042)      | -0.109***<br>(0.036)  | 0.048<br>(0.042)      | 0.069<br>(0.058)       | -0.333***<br>(0.084)   | 0.069<br>(0.058)       | -0.099**<br>(0.048)    | -0.322***<br>(0.075)  | -0.098**<br>(0.048)    |
| $1 - F(c)$                        | -82.397***<br>(6.063) | 38.250***<br>(5.465)  | -82.157***<br>(6.045) | -125.047***<br>(7.612) | 43.017***<br>(5.755)   | -124.683***<br>(7.586) | -149.091***<br>(9.834) | 38.581***<br>(6.547)  | -148.748***<br>(9.808) |
| $(1 - F(c)) \times \Delta y^{nl}$ | 0.0126<br>(0.0109)    | -0.0090<br>(0.0144)   | 0.0126<br>(0.0109)    | -0.0137***<br>(0.0052) | -0.0221***<br>(0.0054) | -0.0137***<br>(0.0052) | -0.0062<br>(0.0119)    | 0.0158<br>(0.0204)    | -0.0061<br>(0.0119)    |
| Constant                          | 42.952***<br>(6.534)  | 0.483<br>(3.082)      | 42.958***<br>(6.535)  | 57.056***<br>(6.167)   | 4.707<br>(7.268)       | 57.061***<br>(6.167)   | 74.894***<br>(5.646)   | 10.386*<br>(5.951)    | 74.919***<br>(5.648)   |
| $N$                               | 3635                  | 3635                  | 3635                  | 3243                   | 3243                   | 3243                   | 3677                   | 3677                  | 3677                   |
| $R^2$                             | 0.419                 | 0.184                 | 0.419                 | 0.442                  | 0.139                  | 0.442                  | 0.522                  | 0.166                 | 0.523                  |

*Notes:* Dependent variable is  $\Delta c_{it}$  (first difference of real total expenditure per adult unit). “Orig.” uses the contemporaneous relative position  $1 - F(c_{it})$ ; “Lagged” uses  $1 - F(c_{i,t-1})$ ; “LOO” uses the leave-one-out ECDF that excludes household  $i$ ’s own observation. Standard errors (in parentheses) are clustered at the household level. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

## Appendix F Consumption Rank versus Income Rank: Additional Details

This appendix supplements the comparison of consumption rank and income rank presented in Section 3.

**Magnitude differences.** The consumption-rank coefficients (Table 2) are larger in magnitude ( $-82$  to  $-149$ ) than the income-rank coefficients (Table 3:  $-54$  to  $-93$ ). This difference is expected: consumption rank  $RP^c = 1 - F(c_{it})$  is mechanically negatively correlated with  $\Delta c_{it}$ ,

inflating the coefficient. Income rank  $RP^y = 1 - F_y(y_{it})$ , constructed from a different variable, is free of this mechanical component.

**Interaction terms.** The interaction  $RP \times \Delta y^{nl}$  is not significant with income rank in any village specification ( $p > 0.30$ ). With consumption rank, it reaches significance in Shirapur ( $p < 0.01$ ). This difference likely reflects the mechanical component: when  $RP^c$  is correlated with  $\Delta c$  by construction, the interaction with  $\Delta y^{nl}$  may pick up spurious heterogeneity in the mechanical relationship rather than genuine differential shock pass-through.

**Robustness across consumption categories.** Appendix A reports consumption-rank results across different consumption aggregates and testing methods. The finding that rank is significant at the 1% level is robust across all specifications.

## Appendix G Out-of-Sample Robustness: ICRISAT VLS2 (2001–2004)

My main analysis uses data from the first wave of the ICRISAT Village Level Studies (VLS1, 1975–1984), following (Townsend, 1994), (Ligon et al., 2002), and (Bold and Broer, 2021). The VLS1 data were collected at approximately fortnightly intervals by resident investigators, yielding high-frequency consumption and income records that are aggregated to the crop-year level.

ICRISAT resurveyed the same villages beginning in 2001 (VLS2), with a larger sample ( $\sim 75$ –100 households per village) and additional modules on migration and gender. (Morten, 2019) uses the VLS2 data (2001–2004) to study the interaction between temporary migration and risk sharing. I use her publicly available replication data to check whether my reduced-form findings and key data moments are robust to this out-of-sample exercise.

While VLS2 data collection continued beyond 2004 (through 2008 and, in some villages, into the 2010s), I restrict attention to the 2001–2004 wave because later waves present substantial merging difficulties. (Morten, 2019) documents that attrition rates increase in later rounds and that tracking households across waves requires matching on names and household identifiers that are not always consistent across survey rounds. The 2001–2004 panel is the most complete and internally consistent segment of the VLS2 data.

Important differences between VLS1 and VLS2 constrain the comparison and explain why I, like the existing literature, base structural estimation on VLS1:

- (i) *Frequency.* VLS2 (2001–2004) was collected via annual recall interviews, not high-frequency resident-investigator visits. This introduces greater measurement error in consumption and income. (Badiani et al., 2013) show that the annual recall introduces random but not systematic error; however, classical measurement error attenuates the income coefficient  $m_2$  toward zero and inflates the variance ratio  $m_1$ .

- (ii) *Panel length.* VLS2 provides only 4 annual observations (3 first differences per household), compared to VLS1's 10 crop years. This makes higher-order moments ( $m_3$ – $m_5$ ) noisier and renders  $m_7$  (subperiod persistence) essentially unidentifiable.
- (iii) *Changed economic environment.* Between 1984 and 2001, these villages underwent dramatic transformation: poverty fell from  $\sim 76\%$  to  $\sim 22\%$ , temporary migration became pervasive ( $\sim 20\%$  of households per year), and government safety nets (NREGA) were introduced. Structural parameters governing informal insurance may differ across these regimes.

Despite these caveats, two findings from VLS2 reinforce my main results.

## Reduced-Form Test on VLS2

Table G.1 replicates my reduced-form risk-sharing test on VLS2 data using consumption rank, following the Morten (2019) variable definitions. I regress the deviation of household consumption growth from the village mean on household income growth, relative position ( $RP_{vc}$ ), and their interaction, with year fixed effects.

**Table G.1**

Risk-Sharing Test with Relative Position: VLS2 (2001–2004)

| Village   | Standard Test       | Approach 1          |                      | Approach 2          |                      | N   |
|-----------|---------------------|---------------------|----------------------|---------------------|----------------------|-----|
|           | Coeff. on<br>Income | Coeff. on<br>Income | Relative<br>Position | Coeff. on<br>Income | Relative<br>Position |     |
| Aurepalle | −0.081*<br>(0.049)  | −0.044<br>(0.042)   | 1.073***<br>(0.127)  | −0.097<br>(0.073)   | 1.062***<br>(0.125)  | 224 |
| Shirapur  | 0.049<br>(0.039)    | 0.029<br>(0.034)    | 1.092***<br>(0.147)  | −0.018<br>(0.057)   | 1.095***<br>(0.147)  | 206 |
| Kanzara   | 0.031<br>(0.035)    | 0.007<br>(0.034)    | 0.530***<br>(0.118)  | 0.005<br>(0.092)    | 0.530***<br>(0.118)  | 127 |

*Notes:* OLS regressions of  $\Delta \log c_{it} - \Delta \log \bar{c}_t$  on  $\Delta \log y_{it}$  with year fixed effects. Robust standard errors in parentheses. \* $p < 0.1$ ; \*\* $p < 0.05$ ; \*\*\* $p < 0.01$ . Data: ICRISAT VLS2 (2001–2004) from (Morten, 2019). Per capita consumption and income in adult-equivalent terms, winsorized at the 1% level by village.

The core finding replicates: relative position is significant at the 1% level in all three villages, with large positive coefficients ( $RP_{vc} = 0.53$ – $1.10$ ). Under Approach 2, the coefficient on income becomes insignificant in all three villages, mirroring the VLS1 result that controlling for relative position eliminates rejections of efficient risk sharing. At the caste level, relative position is significant at the 1% level in 3 of 5 testable caste groups (those with  $\geq 15$  observations), and the income coefficient under Approach 2 is insignificant in all 5 castes (0/5 rejections at 10%), compared to 2/5 rejections in the standard test.

## Data Moments: VLS1 vs. VLS2

Table G.2 compares the seven data moments across the two waves.<sup>7</sup>

**Table G.2**

Data Moments: VLS1 (1975–1984) vs. VLS2 (2001–2004)

| Moment                                              | Aurepalle |        | Shirapur |       | Kanzara |        |
|-----------------------------------------------------|-----------|--------|----------|-------|---------|--------|
|                                                     | VLS1      | VLS2   | VLS1     | VLS2  | VLS1    | VLS2   |
| $m_1$ : $\text{Var}(\Delta c)/\text{Var}(\Delta y)$ | 0.304     | 0.494  | 0.562    | 0.394 | 0.329   | 0.239  |
| $m_2$ : $\beta(\Delta c, \Delta y)$                 | 0.206     | −0.071 | 0.222    | 0.039 | 0.169   | 0.063  |
| $m_3$ : Asym. variance ratio                        | −0.041    | 0.214  | −0.171   | 0.048 | −0.143  | 0.036  |
| $m_4$ : Asym. $\beta$                               | −0.413    | 0.196  | −0.037   | 0.266 | −0.053  | 0.041  |
| $m_5$ : Skewness( $\Delta c$ )                      | −0.218    | −0.315 | −0.165   | 0.087 | −0.053  | −0.511 |
| $m_6$ : XS dispersion                               | 0.343     | 0.696  | 0.556    | 0.700 | 0.476   | 0.457  |
| $m_7$ : $\beta$ persistence                         | 0.081     | —      | 0.077    | —     | 0.137   | —      |
| Households                                          | 34        | 94     | 31       | 87    | 37      | 51     |
| Obs. ( $\Delta$ )                                   | —         | 224    | —        | 206   | —       | 127    |

*Notes:* VLS1 moments computed from monthly data aggregated to crop years (10 years); VLS2 moments computed from annual recall data (4 years). All variables in log per-adult-equivalent terms.  $m_7$  is unreliable for VLS2 due to the short panel (only 3 first-differenced observations per household). VLS2 data from (Morten, 2019).

The most informative comparison is for  $m_6$  (cross-sectional consumption dispersion), the moment that drives my model’s improvement over the (Bold and Broer, 2021) baseline. In VLS2,  $m_6$  is 0.70 for Aurepalle (vs. 0.34 in VLS1), 0.70 for Shirapur (vs. 0.56), and 0.46 for Kanzara (vs. 0.48). Cross-sectional dispersion is at least as large in the more recent data, and in two villages it is substantially larger. This reinforces my central finding: the baseline model’s under-prediction of consumption dispersion is a robust feature of the data, not an artifact of the VLS1 sample. I have not re-estimated the structural model on VLS2 data due to the panel length and measurement concerns discussed above; however, the fact that the VLS2 dispersion target is larger than the VLS1 target suggests that a model generating cross-sectional variation would be quantitatively important in the later period as well.

The time-series moments present a mixed picture. The regression coefficient  $m_2$  is lower in VLS2 for Shirapur and Kanzara, consistent with classical measurement error in income attenuating the coefficient toward zero under annual recall. The variance ratio  $m_1$  is higher in VLS2 for Aurepalle ( $m_1 = 0.49$  vs. 0.30), which could reflect either greater measurement error in consumption inflating the numerator, or genuinely less complete insurance in the more recent period with expanded migration opportunities (Morten, 2019). For Shirapur and Kanzara,  $m_1$  is comparable across waves. The asymmetry moments ( $m_3$ ,  $m_4$ ) change sign, likely reflecting noise from the short panel (3 first differences per household) and changed economic conditions rather than a structural shift in the asymmetry of insurance.

<sup>7</sup>The moments in Table G.2 are computed from level-based per-capita consumption following Morten (2019)’s variable definitions, to maintain comparability with her replication data. This differs from Table 6, which reports log per-adult-equivalent moments used as the structural estimation targets. This accounts for the discrepancies in  $m_6$  between Table 6 (0.15–0.18) and the VLS1 column of Table G.2 (0.34–0.56), and similarly for  $m_3$ – $m_5$ . All structural results in Sections 5–6 are estimated against Table 6.